

# 1 Tables for depth $P = 1$

|                    | ER         | HH          | L2F          | RR          |
|--------------------|------------|-------------|--------------|-------------|
| MPS                |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 130 [40–100] | 51 [30–100] |
| Fixed Angle (Aer)* | -          | -           | -            | -           |
| Fixed Angle        | -          | -           | -            | -           |
| Fixed Angle*       | -          | -           | -            | -           |
| Fourier (Aer)      | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144] | 130 [40–100] | 51 [30–100] |
| Interp (Aer)       | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Linear Ramp (Aer)* | -          | -           | -            | -           |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | 1 [106]     | -            | 2 [40–100]  |
| TQA (Aer)*         | -          | -           | -            | -           |
| TQA                | -          | -           | -            | -           |
| TQA*               | -          | -           | -            | -           |
| PP                 |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 130 [40–100] | 51 [30–100] |
| Fixed Angle        | -          | -           | -            | -           |
| Fixed Angle*       | -          | -           | -            | -           |
| Interp             | 20 [40–50] | 31 [39–144] | 130 [40–100] | 51 [30–100] |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | 1 [106]     | -            | 2 [40–100]  |
| TQA                | -          | -           | -            | -           |
| TQA*               | -          | -           | -            | -           |
| SV                 |            |             |              |             |
| Fourier            | 61 [10–20] | 10 [12]     | 699 [10–20]  | 570 [10–20] |
| Fixed Angle        | 55 [10–20] | -           | -            | 289 [10–20] |
| Fixed Angle*       | 55 [10–20] | -           | -            | 273 [10–20] |
| Interp             | 83 [10–20] | 10 [12]     | 805 [10–22]  | 381 [10–20] |
| Linear Ramp*       | -          | -           | -            | -           |
| TQA                | 65 [10–20] | -           | 592 [10–17]  | 280 [10–20] |
| TQA*               | 63 [10–20] | -           | 584 [10–17]  | 60 [10–20]  |
| TS                 | 81 [10–20] | 10 [12]     | 481 [10–22]  | 368 [10–20] |

Table 1: **Number of Instances (num. nodes in brackets) for  $P = 1$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                    | ER             | HH             | L2F            | RR             |
|--------------------|----------------|----------------|----------------|----------------|
| MPS                |                |                |                |                |
| Fourier            | $81.5 \pm 1.2$ | $75.4 \pm 1.2$ | $72.7 \pm 1.7$ | $76.8 \pm 1.6$ |
| Fixed Angle (Aer)* | -              | -              | -              | -              |
| Fixed Angle        | -              | -              | -              | -              |
| Fixed Angle*       | -              | -              | -              | -              |
| Fourier (Aer)      | $81.1 \pm 1.2$ | $75.7 \pm 1.5$ | $71.5 \pm 2.1$ | $76.3 \pm 1.4$ |
| Interp             | $81.5 \pm 1.2$ | $75.4 \pm 1.2$ | $72.7 \pm 1.7$ | $76.8 \pm 1.6$ |
| Interp (Aer)       | $81.1 \pm 1.2$ | $75.7 \pm 1.5$ | $71.5 \pm 2.1$ | $76.3 \pm 1.4$ |
| Linear Ramp (Aer)* | -              | -              | -              | -              |
| Linear Ramp*       | -              | -              | -              | -              |
| Recursive TS       | -              | 74.4           | -              | $75.2 \pm 0.3$ |
| TQA (Aer)*         | -              | -              | -              | -              |
| TQA                | -              | -              | -              | -              |
| TQA*               | -              | -              | -              | -              |
| PP                 |                |                |                |                |
| Fourier            | $81.5 \pm 1.2$ | $75.4 \pm 1.2$ | $72.7 \pm 1.7$ | $76.8 \pm 1.6$ |
| Fixed Angle        | -              | -              | -              | -              |
| Fixed Angle*       | -              | -              | -              | -              |
| Interp             | $81.5 \pm 1.2$ | $75.4 \pm 1.2$ | $72.7 \pm 1.7$ | $76.8 \pm 1.6$ |
| Linear Ramp*       | -              | -              | -              | -              |
| Recursive TS       | -              | 74.4           | -              | $75.2 \pm 0.3$ |
| TQA                | -              | -              | -              | -              |
| TQA*               | -              | -              | -              | -              |
| SV                 |                |                |                |                |
| Fourier            | $78.8 \pm 3.1$ | $79.3 \pm 3.0$ | $74.8 \pm 4.2$ | $82.0 \pm 4.1$ |
| Fixed Angle        | $78.4 \pm 2.6$ | -              | -              | $80.7 \pm 3.1$ |
| Fixed Angle*       | $79.1 \pm 2.9$ | -              | -              | $81.7 \pm 4.3$ |
| Interp             | $78.8 \pm 2.9$ | $79.3 \pm 3.0$ | $74.6 \pm 3.9$ | $81.9 \pm 4.2$ |
| Linear Ramp*       | -              | -              | -              | -              |
| TQA                | $77.6 \pm 2.6$ | -              | $71.0 \pm 5.1$ | $79.0 \pm 2.9$ |
| TQA*               | $78.8 \pm 3.0$ | -              | $74.7 \pm 3.6$ | $78.1 \pm 2.1$ |
| TS                 | $78.8 \pm 2.9$ | $79.3 \pm 3.0$ | $75.0 \pm 4.8$ | $82.0 \pm 4.3$ |

Table 2: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for  $P = 1$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|                    | Approximation Ratio |                |                |                | Num. Instances |             |              |
|--------------------|---------------------|----------------|----------------|----------------|----------------|-------------|--------------|
|                    | ER                  | HH             | L2F            | RR             | ER             | HH          | L2F          |
| MPS                |                     |                |                |                |                |             |              |
| Fourier            | 81.5 $\pm$ 1.2      | 75.4 $\pm$ 1.2 | 72.7 $\pm$ 1.7 | 76.8 $\pm$ 1.6 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Fixed Angle (Aer)* | -                   | -              | -              | -              | -              | -           | -            |
| Fixed Angle        | -                   | -              | -              | -              | -              | -           | -            |
| Fixed Angle*       | -                   | -              | -              | -              | -              | -           | -            |
| Fourier (Aer)      | 81.1 $\pm$ 1.2      | 75.7 $\pm$ 1.5 | 71.5 $\pm$ 2.1 | 76.3 $\pm$ 1.4 | 10 [40]        | 10 [39]     | 10 [40]      |
| Interp             | 81.5 $\pm$ 1.2      | 75.4 $\pm$ 1.2 | 72.7 $\pm$ 1.7 | 76.8 $\pm$ 1.6 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Interp (Aer)       | 81.1 $\pm$ 1.2      | 75.7 $\pm$ 1.5 | 71.5 $\pm$ 2.1 | 76.3 $\pm$ 1.4 | 10 [40]        | 10 [39]     | 10 [40]      |
| Linear Ramp (Aer)* | -                   | -              | -              | -              | -              | -           | -            |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| Recursive TS       | -                   | 74.4           | -              | 75.2 $\pm$ 0.3 | -              | 1 [106]     | -            |
| TQA (Aer)*         | -                   | -              | -              | -              | -              | -           | -            |
| TQA                | -                   | -              | -              | -              | -              | -           | -            |
| TQA*               | -                   | -              | -              | -              | -              | -           | -            |
| PP                 |                     |                |                |                |                |             |              |
| Fourier            | 81.5 $\pm$ 1.2      | 75.4 $\pm$ 1.2 | 72.7 $\pm$ 1.7 | 76.8 $\pm$ 1.6 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Fixed Angle        | -                   | -              | -              | -              | -              | -           | -            |
| Fixed Angle*       | -                   | -              | -              | -              | -              | -           | -            |
| Interp             | 81.5 $\pm$ 1.2      | 75.4 $\pm$ 1.2 | 72.7 $\pm$ 1.7 | 76.8 $\pm$ 1.6 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| Recursive TS       | -                   | 74.4           | -              | 75.2 $\pm$ 0.3 | -              | 1 [106]     | -            |
| TQA                | -                   | -              | -              | -              | -              | -           | -            |
| TQA*               | -                   | -              | -              | -              | -              | -           | -            |
| SV                 |                     |                |                |                |                |             |              |
| Fourier            | 78.8 $\pm$ 3.1      | 79.3 $\pm$ 3.0 | 74.8 $\pm$ 4.2 | 82.0 $\pm$ 4.1 | 61 [10–20]     | 10 [12]     | 699 [10–20]  |
| Fixed Angle        | 78.4 $\pm$ 2.6      | -              | -              | 80.7 $\pm$ 3.1 | 55 [10–20]     | -           | -            |
| Fixed Angle*       | 79.1 $\pm$ 2.9      | -              | -              | 81.7 $\pm$ 4.3 | 55 [10–20]     | -           | -            |
| Interp             | 78.8 $\pm$ 2.9      | 79.3 $\pm$ 3.0 | 74.6 $\pm$ 3.9 | 81.9 $\pm$ 4.2 | 83 [10–20]     | 10 [12]     | 805 [10–22]  |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| TQA                | 77.6 $\pm$ 2.6      | -              | 71.0 $\pm$ 5.1 | 79.0 $\pm$ 2.9 | 65 [10–20]     | -           | 592 [10–17]  |
| TQA*               | 78.8 $\pm$ 3.0      | -              | 74.7 $\pm$ 3.6 | 78.1 $\pm$ 2.1 | 63 [10–20]     | -           | 584 [10–17]  |
| TS                 | 78.8 $\pm$ 2.9      | 79.3 $\pm$ 3.0 | 75.0 $\pm$ 4.8 | 82.0 $\pm$ 4.3 | 81 [10–20]     | 10 [12]     | 481 [10–22]  |

Table 3: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for  $P = 1$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. Number of instances are shown in green, with the range of graph sizes shown in brackets.

## 2 Tables for depth $P = 2$

|                    | ER         | HH          | L2F          | RR          |
|--------------------|------------|-------------|--------------|-------------|
| MPS                |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 130 [40–100] | 51 [30–100] |
| Fixed Angle (Aer)* | -          | -           | -            | 10 [40]     |
| Fixed Angle        | 10 [50]    | -           | 90 [100]     | 40 [30–70]  |
| Fixed Angle*       | 10 [50]    | -           | 90 [100]     | 40 [30–70]  |
| Fourier (Aer)      | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144] | 130 [40–100] | 51 [30–100] |
| Interp (Aer)       | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Linear Ramp (Aer)* | -          | -           | -            | -           |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA (Aer)*         | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| TQA                | 20 [40–50] | 10 [39]     | 130 [40–100] | 50 [30–70]  |
| TQA*               | 20 [40–50] | 10 [39]     | 130 [40–100] | 50 [30–70]  |
| PP                 |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 130 [40–100] | 51 [30–100] |
| Fixed Angle        | 10 [50]    | -           | 90 [100]     | 40 [30–70]  |
| Fixed Angle*       | 10 [50]    | -           | 90 [100]     | 40 [30–70]  |
| Interp             | 20 [40–50] | 31 [39–144] | 130 [40–100] | 51 [30–100] |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA                | 20 [40–50] | 10 [39]     | 130 [40–100] | 50 [30–70]  |
| TQA*               | 20 [40–50] | 10 [39]     | 130 [40–100] | 50 [30–70]  |
| SV                 |            |             |              |             |
| Fourier            | 61 [10–20] | 10 [12]     | 699 [10–20]  | 567 [10–20] |
| Fixed Angle        | 72 [10–20] | -           | 294 [10–22]  | 370 [10–20] |
| Fixed Angle*       | 72 [10–20] | -           | 293 [10–22]  | 358 [10–20] |
| Interp             | 83 [10–20] | 10 [12]     | 805 [10–22]  | 370 [10–20] |
| Linear Ramp*       | -          | -           | -            | -           |
| TQA                | 83 [10–20] | 10 [12]     | 804 [10–22]  | 359 [10–20] |
| TQA*               | 83 [10–20] | 10 [12]     | 803 [10–22]  | 133 [10–20] |
| TS                 | 81 [10–20] | 10 [12]     | 415 [10–22]  | 353 [10–20] |

Table 4: **Number of Instances (num. nodes in brackets) for  $P = 2$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                    | ER             | HH             | L2F            | RR             |
|--------------------|----------------|----------------|----------------|----------------|
| MPS                |                |                |                |                |
| Fourier            | 70.6 $\pm$ 1.8 | 72.1 $\pm$ 8.3 | 76.1 $\pm$ 4.3 | 75.2 $\pm$ 4.4 |
| Fixed Angle (Aer)* | -              | -              | -              | 79.0 $\pm$ 2.6 |
| Fixed Angle        | 77.2 $\pm$ 0.9 | -              | 79.6 $\pm$ 2.6 | 76.6 $\pm$ 4.8 |
| Fixed Angle*       | 80.4 $\pm$ 1.8 | -              | 79.9 $\pm$ 2.4 | 77.3 $\pm$ 4.4 |
| Fourier (Aer)      | 70.5 $\pm$ 2.6 | 75.7 $\pm$ 4.4 | 70.9 $\pm$ 1.8 | 75.4 $\pm$ 2.2 |
| Interp             | 79.7 $\pm$ 2.0 | 75.9 $\pm$ 6.2 | 79.2 $\pm$ 3.0 | 75.6 $\pm$ 4.0 |
| Interp (Aer)       | 77.5 $\pm$ 1.8 | 82.1 $\pm$ 0.9 | 77.6 $\pm$ 2.1 | 71.1 $\pm$ 2.5 |
| Linear Ramp (Aer)* | -              | -              | -              | -              |
| Linear Ramp*       | -              | -              | -              | -              |
| Recursive TS       | -              | -              | -              | -              |
| TQA (Aer)*         | 75.8 $\pm$ 1.3 | 80.7 $\pm$ 2.2 | 77.1 $\pm$ 2.3 | 77.3 $\pm$ 3.4 |
| TQA                | 69.3 $\pm$ 1.1 | 73.0 $\pm$ 1.7 | 70.0 $\pm$ 3.9 | 72.8 $\pm$ 2.5 |
| TQA*               | 77.4 $\pm$ 1.3 | 83.1 $\pm$ 1.6 | 79.2 $\pm$ 3.0 | 77.6 $\pm$ 4.0 |
| PP                 |                |                |                |                |
| Fourier            | 86.7 $\pm$ 2.1 | 80.8 $\pm$ 2.2 | 79.1 $\pm$ 1.7 | 80.2 $\pm$ 2.5 |
| Fixed Angle        | 86.9 $\pm$ 1.0 | -              | 79.4 $\pm$ 1.8 | 83.6 $\pm$ 1.4 |
| Fixed Angle*       | 87.8 $\pm$ 1.0 | -              | 79.4 $\pm$ 1.8 | 83.6 $\pm$ 1.4 |
| Interp             | 87.4 $\pm$ 1.2 | 82.7 $\pm$ 1.2 | 79.6 $\pm$ 2.0 | 83.8 $\pm$ 1.4 |
| Linear Ramp*       | -              | -              | -              | -              |
| Recursive TS       | -              | -              | -              | -              |
| TQA                | 78.9 $\pm$ 1.3 | 73.0 $\pm$ 1.7 | 70.6 $\pm$ 3.0 | 76.9 $\pm$ 1.9 |
| TQA*               | 87.4 $\pm$ 1.2 | 83.1 $\pm$ 1.6 | 79.6 $\pm$ 2.0 | 83.8 $\pm$ 1.4 |
| SV                 |                |                |                |                |
| Fourier            | 75.8 $\pm$ 9.1 | 77.0 $\pm$ 3.9 | 76.4 $\pm$ 5.8 | 80.6 $\pm$ 8.9 |
| Fixed Angle        | 84.1 $\pm$ 2.1 | -              | 82.2 $\pm$ 2.9 | 87.2 $\pm$ 3.0 |
| Fixed Angle*       | 85.5 $\pm$ 2.3 | -              | 82.6 $\pm$ 2.9 | 88.2 $\pm$ 3.3 |
| Interp             | 85.3 $\pm$ 2.4 | 85.5 $\pm$ 1.8 | 82.5 $\pm$ 4.1 | 86.5 $\pm$ 4.6 |
| Linear Ramp*       | -              | -              | -              | -              |
| TQA                | 76.0 $\pm$ 3.2 | 73.0 $\pm$ 5.0 | 72.4 $\pm$ 4.2 | 75.7 $\pm$ 4.4 |
| TQA*               | 85.5 $\pm$ 2.4 | 86.6 $\pm$ 2.5 | 82.8 $\pm$ 3.1 | 87.4 $\pm$ 3.3 |
| TS                 | 85.2 $\pm$ 2.4 | 87.1 $\pm$ 1.9 | 81.3 $\pm$ 2.8 | 86.1 $\pm$ 4.3 |

Table 5: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for  $P = 2$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|                    | Approximation Ratio |                |                |                | Num. Instances |             |              |
|--------------------|---------------------|----------------|----------------|----------------|----------------|-------------|--------------|
|                    | ER                  | HH             | L2F            | RR             | ER             | HH          | L2F          |
| MPS                |                     |                |                |                |                |             |              |
| Fourier            | $70.6 \pm 1.8$      | $72.1 \pm 8.3$ | $76.1 \pm 4.3$ | $75.2 \pm 4.4$ | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Fixed Angle (Aer)* | -                   | -              | -              | $79.0 \pm 2.6$ | -              | -           | -            |
| Fixed Angle        | $77.2 \pm 0.9$      | -              | $79.6 \pm 2.6$ | $76.6 \pm 4.8$ | 10 [50]        | -           | 90 [100]     |
| Fixed Angle*       | $80.4 \pm 1.8$      | -              | $79.9 \pm 2.4$ | $77.3 \pm 4.4$ | 10 [50]        | -           | 90 [100]     |
| Fourier (Aer)      | $70.5 \pm 2.6$      | $75.7 \pm 4.4$ | $70.9 \pm 1.8$ | $75.4 \pm 2.2$ | 10 [40]        | 10 [39]     | 10 [40]      |
| Interp             | $79.7 \pm 2.0$      | $75.9 \pm 6.2$ | $79.2 \pm 3.0$ | $75.6 \pm 4.0$ | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Interp (Aer)       | $77.5 \pm 1.8$      | $82.1 \pm 0.9$ | $77.6 \pm 2.1$ | $71.1 \pm 2.5$ | 10 [40]        | 10 [39]     | 10 [40]      |
| Linear Ramp (Aer)* | -                   | -              | -              | -              | -              | -           | -            |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| Recursive TS       | -                   | -              | -              | -              | -              | -           | -            |
| TQA (Aer)*         | $75.8 \pm 1.3$      | $80.7 \pm 2.2$ | $77.1 \pm 2.3$ | $77.3 \pm 3.4$ | 10 [40]        | 10 [39]     | 10 [40]      |
| TQA                | $69.3 \pm 1.1$      | $73.0 \pm 1.7$ | $70.0 \pm 3.9$ | $72.8 \pm 2.5$ | 20 [40–50]     | 10 [39]     | 130 [40–100] |
| TQA*               | $77.4 \pm 1.3$      | $83.1 \pm 1.6$ | $79.2 \pm 3.0$ | $77.6 \pm 4.0$ | 20 [40–50]     | 10 [39]     | 130 [40–100] |
| PP                 |                     |                |                |                |                |             |              |
| Fourier            | $86.7 \pm 2.1$      | $80.8 \pm 2.2$ | $79.1 \pm 1.7$ | $80.2 \pm 2.5$ | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Fixed Angle        | $86.9 \pm 1.0$      | -              | $79.4 \pm 1.8$ | $83.6 \pm 1.4$ | 10 [50]        | -           | 90 [100]     |
| Fixed Angle*       | $87.8 \pm 1.0$      | -              | $79.4 \pm 1.8$ | $83.6 \pm 1.4$ | 10 [50]        | -           | 90 [100]     |
| Interp             | $87.4 \pm 1.2$      | $82.7 \pm 1.2$ | $79.6 \pm 2.0$ | $83.8 \pm 1.4$ | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| Recursive TS       | -                   | -              | -              | -              | -              | -           | -            |
| TQA                | $78.9 \pm 1.3$      | $73.0 \pm 1.7$ | $70.6 \pm 3.0$ | $76.9 \pm 1.9$ | 20 [40–50]     | 10 [39]     | 130 [40–100] |
| TQA*               | $87.4 \pm 1.2$      | $83.1 \pm 1.6$ | $79.6 \pm 2.0$ | $83.8 \pm 1.4$ | 20 [40–50]     | 10 [39]     | 130 [40–100] |
| SV                 |                     |                |                |                |                |             |              |
| Fourier            | $75.8 \pm 9.1$      | $77.0 \pm 3.9$ | $76.4 \pm 5.8$ | $80.6 \pm 8.9$ | 61 [10–20]     | 10 [12]     | 699 [10–20]  |
| Fixed Angle        | $84.1 \pm 2.1$      | -              | $82.2 \pm 2.9$ | $87.2 \pm 3.0$ | 72 [10–20]     | -           | 294 [10–22]  |
| Fixed Angle*       | $85.5 \pm 2.3$      | -              | $82.6 \pm 2.9$ | $88.2 \pm 3.3$ | 72 [10–20]     | -           | 293 [10–22]  |
| Interp             | $85.3 \pm 2.4$      | $85.5 \pm 1.8$ | $82.5 \pm 4.1$ | $86.5 \pm 4.6$ | 83 [10–20]     | 10 [12]     | 805 [10–22]  |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| TQA                | $76.0 \pm 3.2$      | $73.0 \pm 5.0$ | $72.4 \pm 4.2$ | $75.7 \pm 4.4$ | 83 [10–20]     | 10 [12]     | 804 [10–22]  |
| TQA*               | $85.5 \pm 2.4$      | $86.6 \pm 2.5$ | $82.8 \pm 3.1$ | $87.4 \pm 3.3$ | 83 [10–20]     | 10 [12]     | 803 [10–22]  |
| TS                 | $85.2 \pm 2.4$      | $87.1 \pm 1.9$ | $81.3 \pm 2.8$ | $86.1 \pm 4.3$ | 81 [10–20]     | 10 [12]     | 415 [10–22]  |

Table 6: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for  $P = 2$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. Number of instances are shown in green, with the range of graph sizes shown in brackets.

### 3 Tables for depth $P = 3$

|                    | ER         | HH          | L2F          | RR          |
|--------------------|------------|-------------|--------------|-------------|
| MPS                |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 130 [40–100] | 41 [30–100] |
| Fixed Angle (Aer)* | -          | -           | -            | -           |
| Fixed Angle        | -          | -           | -            | 30 [30–70]  |
| Fixed Angle*       | -          | -           | -            | 30 [30–70]  |
| Fourier (Aer)      | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144] | 130 [40–100] | 41 [30–100] |
| Interp (Aer)       | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Linear Ramp (Aer)* | -          | -           | -            | -           |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA (Aer)*         | -          | -           | -            | -           |
| TQA                | -          | -           | 10 [40]      | 30 [30–70]  |
| TQA*               | -          | -           | 10 [40]      | 30 [30–70]  |
| PP                 |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 130 [40–100] | 41 [30–100] |
| Fixed Angle        | -          | -           | -            | 30 [30–70]  |
| Fixed Angle*       | -          | -           | -            | 30 [30–70]  |
| Interp             | 20 [40–50] | 31 [39–144] | 130 [40–100] | 41 [30–100] |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA                | -          | -           | 10 [40]      | 30 [30–70]  |
| TQA*               | -          | -           | 10 [40]      | 30 [30–70]  |
| SV                 |            |             |              |             |
| Fourier            | 59 [10–19] | 10 [12]     | 699 [10–20]  | 567 [10–20] |
| Fixed Angle        | 72 [10–20] | -           | 266 [10–20]  | 370 [10–20] |
| Fixed Angle*       | 72 [10–20] | -           | 266 [10–20]  | 362 [10–20] |
| Interp             | 82 [10–20] | 10 [12]     | 803 [10–22]  | 370 [10–20] |
| Linear Ramp*       | -          | -           | -            | -           |
| TQA                | 83 [10–20] | 10 [12]     | 803 [10–22]  | 355 [10–20] |
| TQA*               | 82 [10–20] | 10 [12]     | 803 [10–22]  | 132 [10–20] |
| TS                 | 81 [10–20] | 10 [12]     | 415 [10–22]  | 357 [10–20] |

Table 7: **Number of Instances (num. nodes in brackets) for  $P = 3$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                    | ER             | HH             | L2F            | RR             |
|--------------------|----------------|----------------|----------------|----------------|
| MPS                |                |                |                |                |
| Fourier            | $84.2 \pm 6.6$ | $75.6 \pm 6.8$ | $80.1 \pm 4.4$ | $78.0 \pm 5.3$ |
| Fixed Angle (Aer)* | -              | -              | -              | -              |
| Fixed Angle        | -              | -              | -              | $81.6 \pm 6.2$ |
| Fixed Angle*       | -              | -              | -              | $83.3 \pm 5.2$ |
| Fourier (Aer)      | $80.0 \pm 4.1$ | $79.6 \pm 2.4$ | $75.9 \pm 8.1$ | $77.1 \pm 2.5$ |
| Interp             | $87.9 \pm 1.5$ | $79.0 \pm 7.0$ | $85.7 \pm 1.6$ | $81.4 \pm 6.7$ |
| Interp (Aer)       | $83.4 \pm 3.0$ | $86.1 \pm 1.1$ | $87.9 \pm 1.6$ | $73.4 \pm 3.8$ |
| Linear Ramp (Aer)* | -              | -              | -              | -              |
| Linear Ramp*       | -              | -              | -              | -              |
| Recursive TS       | -              | -              | -              | -              |
| TQA (Aer)*         | -              | -              | -              | -              |
| TQA                | -              | -              | $82.1 \pm 1.8$ | $76.2 \pm 3.9$ |
| TQA*               | -              | -              | $88.8 \pm 1.1$ | $81.5 \pm 6.3$ |
| PP                 |                |                |                |                |
| Fourier            | $86.7 \pm 1.9$ | $83.9 \pm 2.7$ | $81.0 \pm 2.6$ | $84.6 \pm 2.3$ |
| Fixed Angle        | -              | -              | -              | $87.3 \pm 1.3$ |
| Fixed Angle*       | -              | -              | -              | $87.3 \pm 1.3$ |
| Interp             | $89.0 \pm 1.7$ | $86.7 \pm 1.1$ | $83.3 \pm 1.9$ | $87.3 \pm 1.5$ |
| Linear Ramp*       | -              | -              | -              | -              |
| Recursive TS       | -              | -              | -              | -              |
| TQA                | -              | -              | $68.0 \pm 1.6$ | $80.7 \pm 1.4$ |
| TQA*               | -              | -              | $79.7 \pm 2.4$ | $87.3 \pm 1.3$ |
| SV                 |                |                |                |                |
| Fourier            | $83.2 \pm 5.9$ | $79.2 \pm 4.5$ | $78.9 \pm 7.6$ | $88.4 \pm 4.1$ |
| Fixed Angle        | $87.1 \pm 1.8$ | -              | $87.2 \pm 2.5$ | $89.9 \pm 2.2$ |
| Fixed Angle*       | $89.1 \pm 1.9$ | -              | $87.8 \pm 2.5$ | $91.6 \pm 2.7$ |
| Interp             | $88.8 \pm 2.0$ | $89.8 \pm 1.5$ | $87.4 \pm 3.8$ | $89.7 \pm 4.1$ |
| Linear Ramp*       | -              | -              | -              | -              |
| TQA                | $81.1 \pm 4.0$ | $79.7 \pm 4.8$ | $77.3 \pm 5.2$ | $80.1 \pm 5.5$ |
| TQA*               | $87.6 \pm 2.2$ | $88.2 \pm 2.0$ | $85.1 \pm 3.8$ | $89.9 \pm 2.6$ |
| TS                 | $88.0 \pm 2.5$ | $90.1 \pm 1.8$ | $84.9 \pm 2.4$ | $88.5 \pm 3.8$ |

Table 8: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for  $P = 3$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.



|                    | Approximation Ratio |                |                |                | Num. Instances |             |              |
|--------------------|---------------------|----------------|----------------|----------------|----------------|-------------|--------------|
|                    | ER                  | HH             | L2F            | RR             | ER             | HH          | L2F          |
| MPS                |                     |                |                |                |                |             |              |
| Fourier            | 84.2 $\pm$ 6.6      | 75.6 $\pm$ 6.8 | 80.1 $\pm$ 4.4 | 78.0 $\pm$ 5.3 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Fixed Angle (Aer)* | -                   | -              | -              | -              | -              | -           | -            |
| Fixed Angle        | -                   | -              | -              | 81.6 $\pm$ 6.2 | -              | -           | -            |
| Fixed Angle*       | -                   | -              | -              | 83.3 $\pm$ 5.2 | -              | -           | -            |
| Fourier (Aer)      | 80.0 $\pm$ 4.1      | 79.6 $\pm$ 2.4 | 75.9 $\pm$ 8.1 | 77.1 $\pm$ 2.5 | 10 [40]        | 10 [39]     | 10 [40]      |
| Interp             | 87.9 $\pm$ 1.5      | 79.0 $\pm$ 7.0 | 85.7 $\pm$ 1.6 | 81.4 $\pm$ 6.7 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Interp (Aer)       | 83.4 $\pm$ 3.0      | 86.1 $\pm$ 1.1 | 87.9 $\pm$ 1.6 | 73.4 $\pm$ 3.8 | 10 [40]        | 10 [39]     | 10 [40]      |
| Linear Ramp (Aer)* | -                   | -              | -              | -              | -              | -           | -            |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| Recursive TS       | -                   | -              | -              | -              | -              | -           | -            |
| TQA (Aer)*         | -                   | -              | -              | -              | -              | -           | -            |
| TQA                | -                   | -              | 82.1 $\pm$ 1.8 | 76.2 $\pm$ 3.9 | -              | -           | 10 [40]      |
| TQA*               | -                   | -              | 88.8 $\pm$ 1.1 | 81.5 $\pm$ 6.3 | -              | -           | 10 [40]      |
| PP                 |                     |                |                |                |                |             |              |
| Fourier            | 86.7 $\pm$ 1.9      | 83.9 $\pm$ 2.7 | 81.0 $\pm$ 2.6 | 84.6 $\pm$ 2.3 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Fixed Angle        | -                   | -              | -              | 87.3 $\pm$ 1.3 | -              | -           | -            |
| Fixed Angle*       | -                   | -              | -              | 87.3 $\pm$ 1.3 | -              | -           | -            |
| Interp             | 89.0 $\pm$ 1.7      | 86.7 $\pm$ 1.1 | 83.3 $\pm$ 1.9 | 87.3 $\pm$ 1.5 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| Recursive TS       | -                   | -              | -              | -              | -              | -           | -            |
| TQA                | -                   | -              | 68.0 $\pm$ 1.6 | 80.7 $\pm$ 1.4 | -              | -           | 10 [40]      |
| TQA*               | -                   | -              | 79.7 $\pm$ 2.4 | 87.3 $\pm$ 1.3 | -              | -           | 10 [40]      |
| SV                 |                     |                |                |                |                |             |              |
| Fourier            | 83.2 $\pm$ 5.9      | 79.2 $\pm$ 4.5 | 78.9 $\pm$ 7.6 | 88.4 $\pm$ 4.1 | 59 [10–19]     | 10 [12]     | 699 [10–20]  |
| Fixed Angle        | 87.1 $\pm$ 1.8      | -              | 87.2 $\pm$ 2.5 | 89.9 $\pm$ 2.2 | 72 [10–20]     | -           | 266 [10–20]  |
| Fixed Angle*       | 89.1 $\pm$ 1.9      | -              | 87.8 $\pm$ 2.5 | 91.6 $\pm$ 2.7 | 72 [10–20]     | -           | 266 [10–20]  |
| Interp             | 88.8 $\pm$ 2.0      | 89.8 $\pm$ 1.5 | 87.4 $\pm$ 3.8 | 89.7 $\pm$ 4.1 | 82 [10–20]     | 10 [12]     | 803 [10–22]  |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| TQA                | 81.1 $\pm$ 4.0      | 79.7 $\pm$ 4.8 | 77.3 $\pm$ 5.2 | 80.1 $\pm$ 5.5 | 83 [10–20]     | 10 [12]     | 803 [10–22]  |
| TQA*               | 87.6 $\pm$ 2.2      | 88.2 $\pm$ 2.0 | 85.1 $\pm$ 3.8 | 89.9 $\pm$ 2.6 | 82 [10–20]     | 10 [12]     | 803 [10–22]  |
| TS                 | 88.0 $\pm$ 2.5      | 90.1 $\pm$ 1.8 | 84.9 $\pm$ 2.4 | 88.5 $\pm$ 3.8 | 81 [10–20]     | 10 [12]     | 415 [10–22]  |

Table 9: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for  $P = 3$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. Number of instances are shown in green, with the range of graph sizes shown in brackets.

## 4 Tables for depth $P = 4$

|                    | ER         | HH          | L2F          | RR          |
|--------------------|------------|-------------|--------------|-------------|
| MPS                |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 130 [40–100] | 21 [40–100] |
| Fixed Angle (Aer)* | -          | -           | -            | 10 [40]     |
| Fixed Angle        | -          | -           | 30 [100]     | 10 [40]     |
| Fixed Angle*       | -          | -           | 30 [100]     | 10 [40]     |
| Fourier (Aer)      | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144] | 130 [40–100] | 21 [40–100] |
| Interp (Aer)       | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Linear Ramp (Aer)* | -          | -           | -            | -           |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA (Aer)*         | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| TQA                | 20 [40–50] | 10 [39]     | 130 [40–100] | 20 [40]     |
| TQA*               | 20 [40–50] | 10 [39]     | 130 [40–100] | 20 [40]     |
| PP                 |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 130 [40–100] | 21 [40–100] |
| Fixed Angle        | -          | -           | 30 [100]     | 10 [40]     |
| Fixed Angle*       | -          | -           | 30 [100]     | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144] | 130 [40–100] | 21 [40–100] |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA                | 20 [40–50] | 10 [39]     | 130 [40–100] | 20 [40]     |
| TQA*               | 20 [40–50] | 10 [39]     | 130 [40–100] | 20 [40]     |
| SV                 |            |             |              |             |
| Fourier            | 59 [10–19] | 10 [12]     | 699 [10–20]  | 567 [10–20] |
| Fixed Angle        | 52 [10–20] | -           | 95 [10–20]   | 156 [10–20] |
| Fixed Angle*       | 52 [10–20] | -           | 95 [10–20]   | 152 [10–20] |
| Interp             | 82 [10–20] | 10 [12]     | 803 [10–22]  | 370 [10–20] |
| Linear Ramp*       | -          | -           | -            | -           |
| TQA                | 83 [10–20] | 10 [12]     | 803 [10–22]  | 352 [10–20] |
| TQA*               | 82 [10–20] | 10 [12]     | 803 [10–22]  | 133 [10–20] |
| TS                 | 81 [10–20] | 10 [12]     | 414 [10–22]  | 352 [10–20] |

Table 10: **Number of Instances (num. nodes in brackets) for  $P = 4$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                    | ER          | HH         | L2F        | RR         |
|--------------------|-------------|------------|------------|------------|
| MPS                |             |            |            |            |
| Fourier            | 77.4 ± 6.4  | 76.0 ± 8.2 | 83.0 ± 4.4 | 88.0 ± 6.0 |
| Fixed Angle (Aer)* | -           | -          | -          | 93.1 ± 1.0 |
| Fixed Angle        | -           | -          | 90.4 ± 1.5 | 90.4 ± 1.2 |
| Fixed Angle*       | -           | -          | 90.5 ± 1.4 | 92.6 ± 0.8 |
| Fourier (Aer)      | 74.3 ± 4.1  | 82.4 ± 2.6 | 84.8 ± 4.0 | 79.7 ± 3.3 |
| Interp             | 92.1 ± 1.1  | 79.5 ± 8.6 | 89.9 ± 1.1 | 90.5 ± 6.0 |
| Interp (Aer)       | 87.8 ± 2.2  | 88.9 ± 1.1 | 92.4 ± 1.1 | 80.4 ± 6.6 |
| Linear Ramp (Aer)* | -           | -          | -          | -          |
| Linear Ramp*       | -           | -          | -          | -          |
| Recursive TS       | -           | -          | -          | -          |
| TQA (Aer)*         | 80.7 ± 4.4  | 84.8 ± 0.9 | 90.6 ± 1.7 | 82.5 ± 3.4 |
| TQA                | 70.5 ± 3.4  | 82.0 ± 1.6 | 84.2 ± 2.4 | 80.0 ± 1.8 |
| TQA*               | 79.0 ± 7.0  | 86.9 ± 1.1 | 88.5 ± 1.9 | 88.1 ± 3.4 |
| PP                 |             |            |            |            |
| Fourier            | 88.0 ± 2.1  | 86.4 ± 2.7 | 82.4 ± 2.9 | 87.4 ± 1.9 |
| Fixed Angle        | -           | -          | 84.6 ± 1.5 | 89.9 ± 1.6 |
| Fixed Angle*       | -           | -          | 86.5 ± 1.7 | 89.9 ± 1.6 |
| Interp             | 90.7 ± 1.9  | 89.3 ± 1.1 | 85.3 ± 1.8 | 89.3 ± 1.6 |
| Linear Ramp*       | -           | -          | -          | -          |
| Recursive TS       | -           | -          | -          | -          |
| TQA                | 83.7 ± 1.4  | 82.0 ± 1.6 | 75.7 ± 4.8 | 85.4 ± 1.8 |
| TQA*               | 89.2 ± 1.9  | 87.3 ± 1.3 | 83.5 ± 1.9 | 88.0 ± 1.5 |
| SV                 |             |            |            |            |
| Fourier            | 78.3 ± 10.6 | 83.6 ± 3.3 | 80.9 ± 8.0 | 90.3 ± 4.4 |
| Fixed Angle        | 89.0 ± 2.2  | -          | 91.0 ± 2.3 | 92.9 ± 2.2 |
| Fixed Angle*       | 91.5 ± 1.7  | -          | 91.9 ± 2.3 | 93.4 ± 2.2 |
| Interp             | 91.3 ± 1.7  | 91.8 ± 1.5 | 90.4 ± 3.8 | 91.1 ± 4.8 |
| Linear Ramp*       | -           | -          | -          | -          |
| TQA                | 84.6 ± 3.9  | 82.0 ± 4.9 | 79.8 ± 6.0 | 84.0 ± 5.6 |
| TQA*               | 89.4 ± 1.9  | 89.5 ± 1.9 | 87.8 ± 2.7 | 91.1 ± 2.5 |
| TS                 | 90.3 ± 2.2  | 91.4 ± 1.7 | 87.6 ± 2.3 | 90.1 ± 3.5 |

Table 11: **Avg. Approx. Ratio ± standard deviation in percentage for  $P = 4$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|                    | Approximation Ratio |                |                |                | Num. Instances |             |              |
|--------------------|---------------------|----------------|----------------|----------------|----------------|-------------|--------------|
|                    | ER                  | HH             | L2F            | RR             | ER             | HH          | L2F          |
| MPS                |                     |                |                |                |                |             |              |
| Fourier            | 77.4 $\pm$ 6.4      | 76.0 $\pm$ 8.2 | 83.0 $\pm$ 4.4 | 88.0 $\pm$ 6.0 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Fixed Angle (Aer)* | -                   | -              | -              | 93.1 $\pm$ 1.0 | -              | -           | -            |
| Fixed Angle        | -                   | -              | 90.4 $\pm$ 1.5 | 90.4 $\pm$ 1.2 | -              | -           | 30 [100]     |
| Fixed Angle*       | -                   | -              | 90.5 $\pm$ 1.4 | 92.6 $\pm$ 0.8 | -              | -           | 30 [100]     |
| Fourier (Aer)      | 74.3 $\pm$ 4.1      | 82.4 $\pm$ 2.6 | 84.8 $\pm$ 4.0 | 79.7 $\pm$ 3.3 | 10 [40]        | 10 [39]     | 10 [40]      |
| Interp             | 92.1 $\pm$ 1.1      | 79.5 $\pm$ 8.6 | 89.9 $\pm$ 1.1 | 90.5 $\pm$ 6.0 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Interp (Aer)       | 87.8 $\pm$ 2.2      | 88.9 $\pm$ 1.1 | 92.4 $\pm$ 1.1 | 80.4 $\pm$ 6.6 | 10 [40]        | 10 [39]     | 10 [40]      |
| Linear Ramp (Aer)* | -                   | -              | -              | -              | -              | -           | -            |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| Recursive TS       | -                   | -              | -              | -              | -              | -           | -            |
| TQA (Aer)*         | 80.7 $\pm$ 4.4      | 84.8 $\pm$ 0.9 | 90.6 $\pm$ 1.7 | 82.5 $\pm$ 3.4 | 10 [40]        | 10 [39]     | 10 [40]      |
| TQA                | 70.5 $\pm$ 3.4      | 82.0 $\pm$ 1.6 | 84.2 $\pm$ 2.4 | 80.0 $\pm$ 1.8 | 20 [40–50]     | 10 [39]     | 130 [40–100] |
| TQA*               | 79.0 $\pm$ 7.0      | 86.9 $\pm$ 1.1 | 88.5 $\pm$ 1.9 | 88.1 $\pm$ 3.4 | 20 [40–50]     | 10 [39]     | 130 [40–100] |
| PP                 |                     |                |                |                |                |             |              |
| Fourier            | 88.0 $\pm$ 2.1      | 86.4 $\pm$ 2.7 | 82.4 $\pm$ 2.9 | 87.4 $\pm$ 1.9 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Fixed Angle        | -                   | -              | 84.6 $\pm$ 1.5 | 89.9 $\pm$ 1.6 | -              | -           | 30 [100]     |
| Fixed Angle*       | -                   | -              | 86.5 $\pm$ 1.7 | 89.9 $\pm$ 1.6 | -              | -           | 30 [100]     |
| Interp             | 90.7 $\pm$ 1.9      | 89.3 $\pm$ 1.1 | 85.3 $\pm$ 1.8 | 89.3 $\pm$ 1.6 | 20 [40–50]     | 31 [39–144] | 130 [40–100] |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| Recursive TS       | -                   | -              | -              | -              | -              | -           | -            |
| TQA                | 83.7 $\pm$ 1.4      | 82.0 $\pm$ 1.6 | 75.7 $\pm$ 4.8 | 85.4 $\pm$ 1.8 | 20 [40–50]     | 10 [39]     | 130 [40–100] |
| TQA*               | 89.2 $\pm$ 1.9      | 87.3 $\pm$ 1.3 | 83.5 $\pm$ 1.9 | 88.0 $\pm$ 1.5 | 20 [40–50]     | 10 [39]     | 130 [40–100] |
| SV                 |                     |                |                |                |                |             |              |
| Fourier            | 78.3 $\pm$ 10.6     | 83.6 $\pm$ 3.3 | 80.9 $\pm$ 8.0 | 90.3 $\pm$ 4.4 | 59 [10–19]     | 10 [12]     | 699 [10–20]  |
| Fixed Angle        | 89.0 $\pm$ 2.2      | -              | 91.0 $\pm$ 2.3 | 92.9 $\pm$ 2.2 | 52 [10–20]     | -           | 95 [10–20]   |
| Fixed Angle*       | 91.5 $\pm$ 1.7      | -              | 91.9 $\pm$ 2.3 | 93.4 $\pm$ 2.2 | 52 [10–20]     | -           | 95 [10–20]   |
| Interp             | 91.3 $\pm$ 1.7      | 91.8 $\pm$ 1.5 | 90.4 $\pm$ 3.8 | 91.1 $\pm$ 4.8 | 82 [10–20]     | 10 [12]     | 803 [10–22]  |
| Linear Ramp*       | -                   | -              | -              | -              | -              | -           | -            |
| TQA                | 84.6 $\pm$ 3.9      | 82.0 $\pm$ 4.9 | 79.8 $\pm$ 6.0 | 84.0 $\pm$ 5.6 | 83 [10–20]     | 10 [12]     | 803 [10–22]  |
| TQA*               | 89.4 $\pm$ 1.9      | 89.5 $\pm$ 1.9 | 87.8 $\pm$ 2.7 | 91.1 $\pm$ 2.5 | 82 [10–20]     | 10 [12]     | 803 [10–22]  |
| TS                 | 90.3 $\pm$ 2.2      | 91.4 $\pm$ 1.7 | 87.6 $\pm$ 2.3 | 90.1 $\pm$ 3.5 | 81 [10–20]     | 10 [12]     | 414 [10–22]  |

Table 12: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for  $P = 4$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. Number of instances are shown in green, with the range of graph sizes shown in brackets.

## 5 Tables for depth $P = 5$

|                    | ER         | HH           | L2F          | RR          |
|--------------------|------------|--------------|--------------|-------------|
| MPS                |            |              |              |             |
| Fourier            | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 21 [40–100] |
| Fixed Angle (Aer)* | -          | -            | -            | -           |
| Fixed Angle        | -          | -            | -            | 10 [40]     |
| Fixed Angle*       | -          | -            | -            | 10 [40]     |
| Fourier (Aer)      | 10 [40]    | 10 [39]      | 10 [40]      | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 21 [40–100] |
| Interp (Aer)       | 10 [40]    | 10 [39]      | 10 [40]      | 10 [40]     |
| Linear Ramp (Aer)* | -          | -            | -            | -           |
| Linear Ramp*       | -          | -            | -            | -           |
| Recursive TS       | -          | 1 [106]      | -            | 2 [40–100]  |
| TQA (Aer)*         | -          | -            | -            | -           |
| TQA                | -          | 20 [105–144] | 10 [40]      | 10 [40]     |
| TQA*               | -          | 20 [105–144] | 10 [40]      | 10 [40]     |
| PP                 |            |              |              |             |
| Fourier            | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 21 [40–100] |
| Fixed Angle        | -          | -            | -            | 10 [40]     |
| Fixed Angle*       | -          | -            | -            | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 21 [40–100] |
| Linear Ramp*       | -          | -            | -            | -           |
| Recursive TS       | -          | 1 [106]      | -            | 2 [40–100]  |
| TQA                | -          | 20 [105–144] | 10 [40]      | 10 [40]     |
| TQA*               | -          | 20 [105–144] | 10 [40]      | 10 [40]     |
| SV                 |            |              |              |             |
| Fourier            | 58 [10–19] | 10 [12]      | 699 [10–20]  | 260 [10–20] |
| Fixed Angle        | 39 [10–20] | -            | 50 [10–19]   | 119 [10–20] |
| Fixed Angle*       | 39 [10–20] | -            | 49 [10–19]   | 113 [10–20] |
| Interp             | 82 [10–20] | 10 [12]      | 693 [10–20]  | 370 [10–20] |
| Linear Ramp*       | -          | -            | -            | -           |
| TQA                | 83 [10–20] | 10 [12]      | 659 [10–20]  | 357 [10–20] |
| TQA*               | 81 [10–20] | 10 [12]      | 653 [10–20]  | 132 [10–20] |
| TS                 | 81 [10–20] | 10 [12]      | 276 [10–20]  | 348 [10–20] |

Table 13: **Number of Instances (num. nodes in brackets) for  $P = 5$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                    | ER              | HH             | L2F            | RR              |
|--------------------|-----------------|----------------|----------------|-----------------|
| MPS                |                 |                |                |                 |
| Fourier            | 82.0 $\pm$ 7.5  | 77.1 $\pm$ 8.9 | 85.8 $\pm$ 4.6 | 90.0 $\pm$ 8.3  |
| Fixed Angle (Aer)* | -               | -              | -              | -               |
| Fixed Angle        | -               | -              | -              | 93.8 $\pm$ 0.6  |
| Fixed Angle*       | -               | -              | -              | 95.0 $\pm$ 0.5  |
| Fourier (Aer)      | 82.1 $\pm$ 6.3  | 84.7 $\pm$ 2.1 | 90.9 $\pm$ 1.2 | 80.5 $\pm$ 4.8  |
| Interp             | 94.5 $\pm$ 1.2  | 81.5 $\pm$ 8.7 | 92.4 $\pm$ 1.1 | 93.4 $\pm$ 3.6  |
| Interp (Aer)       | 91.6 $\pm$ 1.6  | 90.8 $\pm$ 1.4 | 94.6 $\pm$ 0.8 | 86.3 $\pm$ 4.7  |
| Linear Ramp (Aer)* | -               | -              | -              | -               |
| Linear Ramp*       | -               | -              | -              | -               |
| Recursive TS       | -               | 71.1           | -              | 83.7 $\pm$ 7.3  |
| TQA (Aer)*         | -               | -              | -              | -               |
| TQA                | -               | 73.4 $\pm$ 4.6 | 88.2 $\pm$ 1.2 | 82.6 $\pm$ 1.5  |
| TQA*               | -               | 75.2 $\pm$ 4.8 | 91.4 $\pm$ 1.0 | 90.2 $\pm$ 1.4  |
| PP                 |                 |                |                |                 |
| Fourier            | 89.6 $\pm$ 1.9  | 88.4 $\pm$ 2.7 | 83.5 $\pm$ 2.5 | 89.0 $\pm$ 1.5  |
| Fixed Angle        | -               | -              | -              | 91.5 $\pm$ 1.5  |
| Fixed Angle*       | -               | -              | -              | 91.7 $\pm$ 1.5  |
| Interp             | 91.9 $\pm$ 2.0  | 91.1 $\pm$ 1.0 | 86.6 $\pm$ 1.8 | 90.5 $\pm$ 1.8  |
| Linear Ramp*       | -               | -              | -              | -               |
| Recursive TS       | -               | 88.2           | -              | 89.0 $\pm$ 2.3  |
| TQA                | -               | 83.8 $\pm$ 1.1 | 68.6 $\pm$ 1.7 | 85.9 $\pm$ 1.4  |
| TQA*               | -               | 88.2 $\pm$ 0.8 | 83.5 $\pm$ 2.0 | 89.7 $\pm$ 1.5  |
| SV                 |                 |                |                |                 |
| Fourier            | 80.0 $\pm$ 10.3 | 79.2 $\pm$ 6.8 | 81.4 $\pm$ 8.3 | 85.5 $\pm$ 10.6 |
| Fixed Angle        | 90.0 $\pm$ 2.3  | -              | 94.0 $\pm$ 1.8 | 94.9 $\pm$ 2.0  |
| Fixed Angle*       | 93.2 $\pm$ 1.5  | -              | 95.2 $\pm$ 1.7 | 95.4 $\pm$ 2.0  |
| Interp             | 93.0 $\pm$ 1.5  | 92.5 $\pm$ 0.9 | 92.6 $\pm$ 4.1 | 92.5 $\pm$ 4.4  |
| Linear Ramp*       | -               | -              | -              | -               |
| TQA                | 86.7 $\pm$ 3.7  | 83.7 $\pm$ 4.9 | 81.9 $\pm$ 6.4 | 86.6 $\pm$ 5.3  |
| TQA*               | 91.6 $\pm$ 1.5  | 90.7 $\pm$ 2.0 | 90.6 $\pm$ 3.4 | 93.2 $\pm$ 2.2  |
| TS                 | 91.9 $\pm$ 2.1  | 92.6 $\pm$ 1.3 | 89.8 $\pm$ 3.5 | 91.1 $\pm$ 3.5  |

Table 14: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for  $P = 5$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|                    | Approximation Ratio |                |                |                 | Num. Instances |              |              |
|--------------------|---------------------|----------------|----------------|-----------------|----------------|--------------|--------------|
|                    | ER                  | HH             | L2F            | RR              | ER             | HH           | L2F          |
| MPS                |                     |                |                |                 |                |              |              |
| Fourier            | 82.0 $\pm$ 7.5      | 77.1 $\pm$ 8.9 | 85.8 $\pm$ 4.6 | 90.0 $\pm$ 8.3  | 20 [40–50]     | 31 [39–144]  | 130 [40–144] |
| Fixed Angle (Aer)* | -                   | -              | -              | -               | -              | -            | -            |
| Fixed Angle        | -                   | -              | -              | 93.8 $\pm$ 0.6  | -              | -            | -            |
| Fixed Angle*       | -                   | -              | -              | 95.0 $\pm$ 0.5  | -              | -            | -            |
| Fourier (Aer)      | 82.1 $\pm$ 6.3      | 84.7 $\pm$ 2.1 | 90.9 $\pm$ 1.2 | 80.5 $\pm$ 4.8  | 10 [40]        | 10 [39]      | 10 [40]      |
| Interp             | 94.5 $\pm$ 1.2      | 81.5 $\pm$ 8.7 | 92.4 $\pm$ 1.1 | 93.4 $\pm$ 3.6  | 20 [40–50]     | 31 [39–144]  | 130 [40–144] |
| Interp (Aer)       | 91.6 $\pm$ 1.6      | 90.8 $\pm$ 1.4 | 94.6 $\pm$ 0.8 | 86.3 $\pm$ 4.7  | 10 [40]        | 10 [39]      | 10 [40]      |
| Linear Ramp (Aer)* | -                   | -              | -              | -               | -              | -            | -            |
| Linear Ramp*       | -                   | -              | -              | -               | -              | -            | -            |
| Recursive TS       | -                   | 71.1           | -              | 83.7 $\pm$ 7.3  | -              | 1 [106]      | -            |
| TQA (Aer)*         | -                   | -              | -              | -               | -              | -            | -            |
| TQA                | -                   | 73.4 $\pm$ 4.6 | 88.2 $\pm$ 1.2 | 82.6 $\pm$ 1.5  | -              | 20 [105–144] | 10 [40]      |
| TQA*               | -                   | 75.2 $\pm$ 4.8 | 91.4 $\pm$ 1.0 | 90.2 $\pm$ 1.4  | -              | 20 [105–144] | 10 [40]      |
| PP                 |                     |                |                |                 |                |              |              |
| Fourier            | 89.6 $\pm$ 1.9      | 88.4 $\pm$ 2.7 | 83.5 $\pm$ 2.5 | 89.0 $\pm$ 1.5  | 20 [40–50]     | 31 [39–144]  | 130 [40–144] |
| Fixed Angle        | -                   | -              | -              | 91.5 $\pm$ 1.5  | -              | -            | -            |
| Fixed Angle*       | -                   | -              | -              | 91.7 $\pm$ 1.5  | -              | -            | -            |
| Interp             | 91.9 $\pm$ 2.0      | 91.1 $\pm$ 1.0 | 86.6 $\pm$ 1.8 | 90.5 $\pm$ 1.8  | 20 [40–50]     | 31 [39–144]  | 130 [40–144] |
| Linear Ramp*       | -                   | -              | -              | -               | -              | -            | -            |
| Recursive TS       | -                   | 88.2           | -              | 89.0 $\pm$ 2.3  | -              | 1 [106]      | -            |
| TQA                | -                   | 83.8 $\pm$ 1.1 | 68.6 $\pm$ 1.7 | 85.9 $\pm$ 1.4  | -              | 20 [105–144] | 10 [40]      |
| TQA*               | -                   | 88.2 $\pm$ 0.8 | 83.5 $\pm$ 2.0 | 89.7 $\pm$ 1.5  | -              | 20 [105–144] | 10 [40]      |
| SV                 |                     |                |                |                 |                |              |              |
| Fourier            | 80.0 $\pm$ 10.3     | 79.2 $\pm$ 6.8 | 81.4 $\pm$ 8.3 | 85.5 $\pm$ 10.6 | 58 [10–19]     | 10 [12]      | 699 [10–19]  |
| Fixed Angle        | 90.0 $\pm$ 2.3      | -              | 94.0 $\pm$ 1.8 | 94.9 $\pm$ 2.0  | 39 [10–20]     | -            | 50 [10–19]   |
| Fixed Angle*       | 93.2 $\pm$ 1.5      | -              | 95.2 $\pm$ 1.7 | 95.4 $\pm$ 2.0  | 39 [10–20]     | -            | 49 [10–19]   |
| Interp             | 93.0 $\pm$ 1.5      | 92.5 $\pm$ 0.9 | 92.6 $\pm$ 4.1 | 92.5 $\pm$ 4.4  | 82 [10–20]     | 10 [12]      | 693 [10–19]  |
| Linear Ramp*       | -                   | -              | -              | -               | -              | -            | -            |
| TQA                | 86.7 $\pm$ 3.7      | 83.7 $\pm$ 4.9 | 81.9 $\pm$ 6.4 | 86.6 $\pm$ 5.3  | 83 [10–20]     | 10 [12]      | 659 [10–19]  |
| TQA*               | 91.6 $\pm$ 1.5      | 90.7 $\pm$ 2.0 | 90.6 $\pm$ 3.4 | 93.2 $\pm$ 2.2  | 81 [10–20]     | 10 [12]      | 653 [10–19]  |
| TS                 | 91.9 $\pm$ 2.1      | 92.6 $\pm$ 1.3 | 89.8 $\pm$ 3.5 | 91.1 $\pm$ 3.5  | 81 [10–20]     | 10 [12]      | 276 [10–19]  |

Table 15: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for  $P = 5$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. Number of instances are shown in green, with the range of graph sizes shown in brackets.

## 6 Tables for depth $P = 6$

|                    | ER         | HH          | L2F          | RR          |
|--------------------|------------|-------------|--------------|-------------|
| MPS                |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 130 [40–100] | 21 [40–100] |
| Fixed Angle (Aer)* | -          | -           | -            | 10 [40]     |
| Fixed Angle        | -          | -           | 10 [100]     | 10 [40]     |
| Fixed Angle*       | -          | -           | 10 [100]     | 10 [40]     |
| Fourier (Aer)      | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144] | 130 [40–100] | 21 [40–100] |
| Interp (Aer)       | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Linear Ramp (Aer)* | -          | -           | -            | -           |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA (Aer)*         | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| TQA                | 20 [40–50] | 10 [39]     | 130 [40–100] | 20 [40]     |
| TQA*               | 20 [40–50] | 10 [39]     | 130 [40–100] | 20 [40]     |
| PP                 |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 130 [40–100] | 21 [40–100] |
| Fixed Angle        | -          | -           | 10 [100]     | 10 [40]     |
| Fixed Angle*       | -          | -           | 10 [100]     | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144] | 130 [40–100] | 21 [40–100] |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA                | 20 [40–50] | 10 [39]     | 130 [40–100] | 20 [40]     |
| TQA*               | 20 [40–50] | 10 [39]     | 130 [40–100] | 20 [40]     |
| SV                 |            |             |              |             |
| Fourier            | 56 [10–19] | 10 [12]     | 693 [10–20]  | 260 [10–20] |
| Fixed Angle        | 16 [11–17] | -           | 19 [10–18]   | 59 [10–20]  |
| Fixed Angle*       | 16 [11–17] | -           | 19 [10–18]   | 57 [10–20]  |
| Interp             | 82 [10–20] | 10 [12]     | 686 [10–20]  | 370 [10–20] |
| Linear Ramp*       | -          | -           | -            | -           |
| TQA                | 83 [10–20] | 10 [12]     | 659 [10–20]  | 358 [10–20] |
| TQA*               | 81 [10–20] | 10 [12]     | 653 [10–20]  | 132 [10–20] |
| TS                 | 80 [10–20] | 10 [12]     | 269 [10–20]  | 348 [10–20] |

Table 16: **Number of Instances (num. nodes in brackets) for  $P = 6$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.



|                    | ER          | HH          | L2F        | RR          |
|--------------------|-------------|-------------|------------|-------------|
| MPS                |             |             |            |             |
| Fourier            | 78.9 ± 7.6  | 76.6 ± 10.0 | 87.9 ± 3.5 | 92.2 ± 6.7  |
| Fixed Angle (Aer)* | -           | -           | -          | 97.2 ± 0.2  |
| Fixed Angle        | -           | -           | 95.1 ± 1.0 | 96.1 ± 0.5  |
| Fixed Angle*       | -           | -           | 95.1 ± 1.1 | 96.9 ± 0.6  |
| Fourier (Aer)      | 79.3 ± 7.1  | 85.1 ± 2.5  | 90.3 ± 2.8 | 86.1 ± 3.2  |
| Interp             | 95.8 ± 1.1  | 81.9 ± 9.7  | 93.9 ± 1.0 | 94.3 ± 3.6  |
| Interp (Aer)       | 92.8 ± 1.3  | 92.4 ± 1.2  | 95.9 ± 0.6 | 89.3 ± 3.4  |
| Linear Ramp (Aer)* | -           | -           | -          | -           |
| Linear Ramp*       | -           | -           | -          | -           |
| Recursive TS       | -           | -           | -          | -           |
| TQA (Aer)*         | 83.3 ± 3.8  | 86.8 ± 1.3  | 92.4 ± 1.7 | 93.2 ± 1.1  |
| TQA                | 71.7 ± 6.0  | 85.6 ± 1.4  | 88.0 ± 2.1 | 89.9 ± 2.7  |
| TQA*               | 85.5 ± 6.7  | 88.8 ± 0.9  | 91.1 ± 1.4 | 93.9 ± 1.3  |
| PP                 |             |             |            |             |
| Fourier            | 90.7 ± 1.9  | 89.3 ± 2.8  | 84.8 ± 2.2 | 89.7 ± 1.8  |
| Fixed Angle        | -           | -           | 85.5 ± 1.6 | 92.2 ± 1.6  |
| Fixed Angle*       | -           | -           | 89.1 ± 1.8 | 92.5 ± 1.5  |
| Interp             | 92.7 ± 2.1  | 92.2 ± 0.9  | 87.6 ± 1.8 | 91.6 ± 1.9  |
| Linear Ramp*       | -           | -           | -          | -           |
| Recursive TS       | -           | -           | -          | -           |
| TQA                | 84.6 ± 1.9  | 85.6 ± 1.4  | 76.5 ± 5.9 | 88.0 ± 1.8  |
| TQA*               | 91.3 ± 2.4  | 89.8 ± 1.0  | 85.2 ± 2.2 | 90.7 ± 1.6  |
| SV                 |             |             |            |             |
| Fourier            | 79.8 ± 11.1 | 81.3 ± 11.1 | 81.1 ± 9.4 | 84.9 ± 11.8 |
| Fixed Angle        | 90.5 ± 2.5  | -           | 95.7 ± 1.3 | 95.9 ± 1.3  |
| Fixed Angle*       | 94.5 ± 1.5  | -           | 96.6 ± 1.4 | 96.3 ± 1.4  |
| Interp             | 94.2 ± 1.3  | 93.6 ± 1.4  | 93.8 ± 4.0 | 93.1 ± 4.9  |
| Linear Ramp*       | -           | -           | -          | -           |
| TQA                | 88.1 ± 3.5  | 85.6 ± 4.4  | 82.4 ± 7.3 | 88.4 ± 5.0  |
| TQA*               | 93.0 ± 1.4  | 91.2 ± 2.2  | 91.9 ± 3.1 | 94.3 ± 2.2  |
| TS                 | 93.1 ± 1.9  | 93.1 ± 1.3  | 91.0 ± 2.7 | 92.2 ± 3.4  |

Table 17: **Avg. Approx. Ratio ± standard deviation in percentage for  $P = 6$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|                    | Approximation Ratio |                 |                |                 | Num. Instances |             |              |
|--------------------|---------------------|-----------------|----------------|-----------------|----------------|-------------|--------------|
|                    | ER                  | HH              | L2F            | RR              | ER             | HH          | L2F          |
| MPS                |                     |                 |                |                 |                |             |              |
| Fourier            | 78.9 $\pm$ 7.6      | 76.6 $\pm$ 10.0 | 87.9 $\pm$ 3.5 | 92.2 $\pm$ 6.7  | 20 [40–50]     | 31 [39–144] | 130 [40–144] |
| Fixed Angle (Aer)* | -                   | -               | -              | 97.2 $\pm$ 0.2  | -              | -           | -            |
| Fixed Angle        | -                   | -               | 95.1 $\pm$ 1.0 | 96.1 $\pm$ 0.5  | -              | -           | 10 [100–144] |
| Fixed Angle*       | -                   | -               | 95.1 $\pm$ 1.1 | 96.9 $\pm$ 0.6  | -              | -           | 10 [100–144] |
| Fourier (Aer)      | 79.3 $\pm$ 7.1      | 85.1 $\pm$ 2.5  | 90.3 $\pm$ 2.8 | 86.1 $\pm$ 3.2  | 10 [40]        | 10 [39]     | 10 [40]      |
| Interp             | 95.8 $\pm$ 1.1      | 81.9 $\pm$ 9.7  | 93.9 $\pm$ 1.0 | 94.3 $\pm$ 3.6  | 20 [40–50]     | 31 [39–144] | 130 [40–144] |
| Interp (Aer)       | 92.8 $\pm$ 1.3      | 92.4 $\pm$ 1.2  | 95.9 $\pm$ 0.6 | 89.3 $\pm$ 3.4  | 10 [40]        | 10 [39]     | 10 [40]      |
| Linear Ramp (Aer)* | -                   | -               | -              | -               | -              | -           | -            |
| Linear Ramp*       | -                   | -               | -              | -               | -              | -           | -            |
| Recursive TS       | -                   | -               | -              | -               | -              | -           | -            |
| TQA (Aer)*         | 83.3 $\pm$ 3.8      | 86.8 $\pm$ 1.3  | 92.4 $\pm$ 1.7 | 93.2 $\pm$ 1.1  | 10 [40]        | 10 [39]     | 10 [40]      |
| TQA                | 71.7 $\pm$ 6.0      | 85.6 $\pm$ 1.4  | 88.0 $\pm$ 2.1 | 89.9 $\pm$ 2.7  | 20 [40–50]     | 10 [39]     | 130 [40–144] |
| TQA*               | 85.5 $\pm$ 6.7      | 88.8 $\pm$ 0.9  | 91.1 $\pm$ 1.4 | 93.9 $\pm$ 1.3  | 20 [40–50]     | 10 [39]     | 130 [40–144] |
| PP                 |                     |                 |                |                 |                |             |              |
| Fourier            | 90.7 $\pm$ 1.9      | 89.3 $\pm$ 2.8  | 84.8 $\pm$ 2.2 | 89.7 $\pm$ 1.8  | 20 [40–50]     | 31 [39–144] | 130 [40–144] |
| Fixed Angle        | -                   | -               | 85.5 $\pm$ 1.6 | 92.2 $\pm$ 1.6  | -              | -           | 10 [100–144] |
| Fixed Angle*       | -                   | -               | 89.1 $\pm$ 1.8 | 92.5 $\pm$ 1.5  | -              | -           | 10 [100–144] |
| Interp             | 92.7 $\pm$ 2.1      | 92.2 $\pm$ 0.9  | 87.6 $\pm$ 1.8 | 91.6 $\pm$ 1.9  | 20 [40–50]     | 31 [39–144] | 130 [40–144] |
| Linear Ramp*       | -                   | -               | -              | -               | -              | -           | -            |
| Recursive TS       | -                   | -               | -              | -               | -              | -           | -            |
| TQA                | 84.6 $\pm$ 1.9      | 85.6 $\pm$ 1.4  | 76.5 $\pm$ 5.9 | 88.0 $\pm$ 1.8  | 20 [40–50]     | 10 [39]     | 130 [40–144] |
| TQA*               | 91.3 $\pm$ 2.4      | 89.8 $\pm$ 1.0  | 85.2 $\pm$ 2.2 | 90.7 $\pm$ 1.6  | 20 [40–50]     | 10 [39]     | 130 [40–144] |
| SV                 |                     |                 |                |                 |                |             |              |
| Fourier            | 79.8 $\pm$ 11.1     | 81.3 $\pm$ 11.1 | 81.1 $\pm$ 9.4 | 84.9 $\pm$ 11.8 | 56 [10–19]     | 10 [12]     | 693 [10–144] |
| Fixed Angle        | 90.5 $\pm$ 2.5      | -               | 95.7 $\pm$ 1.3 | 95.9 $\pm$ 1.3  | 16 [11–17]     | -           | 19 [10–144]  |
| Fixed Angle*       | 94.5 $\pm$ 1.5      | -               | 96.6 $\pm$ 1.4 | 96.3 $\pm$ 1.4  | 16 [11–17]     | -           | 19 [10–144]  |
| Interp             | 94.2 $\pm$ 1.3      | 93.6 $\pm$ 1.4  | 93.8 $\pm$ 4.0 | 93.1 $\pm$ 4.9  | 82 [10–20]     | 10 [12]     | 686 [10–144] |
| Linear Ramp*       | -                   | -               | -              | -               | -              | -           | -            |
| TQA                | 88.1 $\pm$ 3.5      | 85.6 $\pm$ 4.4  | 82.4 $\pm$ 7.3 | 88.4 $\pm$ 5.0  | 83 [10–20]     | 10 [12]     | 659 [10–144] |
| TQA*               | 93.0 $\pm$ 1.4      | 91.2 $\pm$ 2.2  | 91.9 $\pm$ 3.1 | 94.3 $\pm$ 2.2  | 81 [10–20]     | 10 [12]     | 653 [10–144] |
| TS                 | 93.1 $\pm$ 1.9      | 93.1 $\pm$ 1.3  | 91.0 $\pm$ 2.7 | 92.2 $\pm$ 3.4  | 80 [10–20]     | 10 [12]     | 269 [10–144] |

Table 18: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for  $P = 6$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. Number of instances are shown in green, with the range of graph sizes shown in brackets.

## 7 Tables for depth $P = 7$

|                    | ER         | HH          | L2F          | RR          |
|--------------------|------------|-------------|--------------|-------------|
| MPS                |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 120 [40–100] | 21 [40–100] |
| Fixed Angle (Aer)* | -          | -           | -            | -           |
| Fixed Angle        | -          | -           | -            | -           |
| Fixed Angle*       | -          | -           | -            | -           |
| Fourier (Aer)      | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144] | 120 [40–100] | 21 [40–100] |
| Interp (Aer)       | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Linear Ramp (Aer)* | -          | -           | -            | -           |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA (Aer)*         | -          | -           | -            | -           |
| TQA                | -          | -           | -            | -           |
| TQA*               | -          | -           | -            | -           |
| PP                 |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 120 [40–100] | 21 [40–100] |
| Fixed Angle        | -          | -           | -            | -           |
| Fixed Angle*       | -          | -           | -            | -           |
| Interp             | 20 [40–50] | 31 [39–144] | 120 [40–100] | 21 [40–100] |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA                | -          | -           | -            | -           |
| TQA*               | -          | -           | -            | -           |
| SV                 |            |             |              |             |
| Fourier            | 54 [10–18] | 10 [12]     | 688 [10–20]  | 260 [10–20] |
| Fixed Angle        | 16 [11–17] | -           | 18 [10–18]   | 60 [10–20]  |
| Fixed Angle*       | 16 [11–17] | -           | 18 [10–18]   | 57 [10–20]  |
| Interp             | 82 [10–20] | 10 [12]     | 681 [10–20]  | 370 [10–20] |
| Linear Ramp*       | -          | -           | -            | -           |
| TQA                | 83 [10–20] | 10 [12]     | 659 [10–20]  | 358 [10–20] |
| TQA*               | 81 [10–20] | 10 [12]     | 653 [10–20]  | 131 [10–20] |
| TS                 | 80 [10–20] | 10 [12]     | 270 [10–20]  | 349 [10–20] |

Table 19: **Number of Instances (num. nodes in brackets) for  $P = 7$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                    | ER              | HH              | L2F            | RR              |
|--------------------|-----------------|-----------------|----------------|-----------------|
| MPS                |                 |                 |                |                 |
| Fourier            | 81.1 $\pm$ 7.1  | 77.6 $\pm$ 10.1 | 88.1 $\pm$ 4.1 | 91.0 $\pm$ 7.6  |
| Fixed Angle (Aer)* | -               | -               | -              | -               |
| Fixed Angle        | -               | -               | -              | -               |
| Fixed Angle*       | -               | -               | -              | -               |
| Fourier (Aer)      | 81.3 $\pm$ 6.9  | 86.4 $\pm$ 3.1  | 90.1 $\pm$ 3.7 | 88.2 $\pm$ 6.1  |
| Interp             | 96.4 $\pm$ 1.3  | 83.7 $\pm$ 9.4  | 94.9 $\pm$ 1.0 | 95.1 $\pm$ 3.9  |
| Interp (Aer)       | 94.0 $\pm$ 1.5  | 93.6 $\pm$ 1.0  | 96.7 $\pm$ 0.6 | 90.4 $\pm$ 2.4  |
| Linear Ramp (Aer)* | -               | -               | -              | -               |
| Linear Ramp*       | -               | -               | -              | -               |
| Recursive TS       | -               | -               | -              | -               |
| TQA (Aer)*         | -               | -               | -              | -               |
| TQA                | -               | -               | -              | -               |
| TQA*               | -               | -               | -              | -               |
| PP                 |                 |                 |                |                 |
| Fourier            | 90.0 $\pm$ 2.8  | 89.9 $\pm$ 2.7  | 85.2 $\pm$ 2.1 | 90.1 $\pm$ 1.8  |
| Fixed Angle        | -               | -               | -              | -               |
| Fixed Angle*       | -               | -               | -              | -               |
| Interp             | 93.3 $\pm$ 2.1  | 93.1 $\pm$ 0.9  | 88.3 $\pm$ 1.8 | 91.5 $\pm$ 2.0  |
| Linear Ramp*       | -               | -               | -              | -               |
| Recursive TS       | -               | -               | -              | -               |
| TQA                | -               | -               | -              | -               |
| TQA*               | -               | -               | -              | -               |
| SV                 |                 |                 |                |                 |
| Fourier            | 80.0 $\pm$ 10.9 | 81.7 $\pm$ 10.2 | 81.0 $\pm$ 9.8 | 84.9 $\pm$ 12.2 |
| Fixed Angle        | 91.0 $\pm$ 3.0  | -               | 96.4 $\pm$ 1.0 | 96.8 $\pm$ 1.2  |
| Fixed Angle*       | 95.5 $\pm$ 1.6  | -               | 97.5 $\pm$ 1.1 | 97.2 $\pm$ 1.2  |
| Interp             | 95.1 $\pm$ 1.3  | 94.3 $\pm$ 1.2  | 94.7 $\pm$ 4.2 | 93.3 $\pm$ 5.4  |
| Linear Ramp*       | -               | -               | -              | -               |
| TQA                | 89.2 $\pm$ 3.4  | 87.0 $\pm$ 4.1  | 83.0 $\pm$ 7.8 | 89.3 $\pm$ 4.8  |
| TQA*               | 93.9 $\pm$ 1.4  | 91.8 $\pm$ 2.2  | 92.5 $\pm$ 3.2 | 95.1 $\pm$ 1.8  |
| TS                 | 93.9 $\pm$ 1.9  | 93.8 $\pm$ 1.0  | 92.4 $\pm$ 4.1 | 92.9 $\pm$ 3.4  |

Table 20: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for  $P = 7$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|                    | Approximation Ratio |                 |                |                 | Num. Instances |             |              |
|--------------------|---------------------|-----------------|----------------|-----------------|----------------|-------------|--------------|
|                    | ER                  | HH              | L2F            | RR              | ER             | HH          | L2F          |
| MPS                |                     |                 |                |                 |                |             |              |
| Fourier            | 81.1 $\pm$ 7.1      | 77.6 $\pm$ 10.1 | 88.1 $\pm$ 4.1 | 91.0 $\pm$ 7.6  | 20 [40–50]     | 31 [39–144] | 120 [40–144] |
| Fixed Angle (Aer)* | -                   | -               | -              | -               | -              | -           | -            |
| Fixed Angle        | -                   | -               | -              | -               | -              | -           | -            |
| Fixed Angle*       | -                   | -               | -              | -               | -              | -           | -            |
| Fourier (Aer)      | 81.3 $\pm$ 6.9      | 86.4 $\pm$ 3.1  | 90.1 $\pm$ 3.7 | 88.2 $\pm$ 6.1  | 10 [40]        | 10 [39]     | 10 [40]      |
| Interp             | 96.4 $\pm$ 1.3      | 83.7 $\pm$ 9.4  | 94.9 $\pm$ 1.0 | 95.1 $\pm$ 3.9  | 20 [40–50]     | 31 [39–144] | 120 [40–144] |
| Interp (Aer)       | 94.0 $\pm$ 1.5      | 93.6 $\pm$ 1.0  | 96.7 $\pm$ 0.6 | 90.4 $\pm$ 2.4  | 10 [40]        | 10 [39]     | 10 [40]      |
| Linear Ramp (Aer)* | -                   | -               | -              | -               | -              | -           | -            |
| Linear Ramp*       | -                   | -               | -              | -               | -              | -           | -            |
| Recursive TS       | -                   | -               | -              | -               | -              | -           | -            |
| TQA (Aer)*         | -                   | -               | -              | -               | -              | -           | -            |
| TQA                | -                   | -               | -              | -               | -              | -           | -            |
| TQA*               | -                   | -               | -              | -               | -              | -           | -            |
| PP                 |                     |                 |                |                 |                |             |              |
| Fourier            | 90.0 $\pm$ 2.8      | 89.9 $\pm$ 2.7  | 85.2 $\pm$ 2.1 | 90.1 $\pm$ 1.8  | 20 [40–50]     | 31 [39–144] | 120 [40–144] |
| Fixed Angle        | -                   | -               | -              | -               | -              | -           | -            |
| Fixed Angle*       | -                   | -               | -              | -               | -              | -           | -            |
| Interp             | 93.3 $\pm$ 2.1      | 93.1 $\pm$ 0.9  | 88.3 $\pm$ 1.8 | 91.5 $\pm$ 2.0  | 20 [40–50]     | 31 [39–144] | 120 [40–144] |
| Linear Ramp*       | -                   | -               | -              | -               | -              | -           | -            |
| Recursive TS       | -                   | -               | -              | -               | -              | -           | -            |
| TQA                | -                   | -               | -              | -               | -              | -           | -            |
| TQA*               | -                   | -               | -              | -               | -              | -           | -            |
| SV                 |                     |                 |                |                 |                |             |              |
| Fourier            | 80.0 $\pm$ 10.9     | 81.7 $\pm$ 10.2 | 81.0 $\pm$ 9.8 | 84.9 $\pm$ 12.2 | 54 [10–18]     | 10 [12]     | 688 [10–144] |
| Fixed Angle        | 91.0 $\pm$ 3.0      | -               | 96.4 $\pm$ 1.0 | 96.8 $\pm$ 1.2  | 16 [11–17]     | -           | 18 [10–144]  |
| Fixed Angle*       | 95.5 $\pm$ 1.6      | -               | 97.5 $\pm$ 1.1 | 97.2 $\pm$ 1.2  | 16 [11–17]     | -           | 18 [10–144]  |
| Interp             | 95.1 $\pm$ 1.3      | 94.3 $\pm$ 1.2  | 94.7 $\pm$ 4.2 | 93.3 $\pm$ 5.4  | 82 [10–20]     | 10 [12]     | 681 [10–144] |
| Linear Ramp*       | -                   | -               | -              | -               | -              | -           | -            |
| TQA                | 89.2 $\pm$ 3.4      | 87.0 $\pm$ 4.1  | 83.0 $\pm$ 7.8 | 89.3 $\pm$ 4.8  | 83 [10–20]     | 10 [12]     | 659 [10–144] |
| TQA*               | 93.9 $\pm$ 1.4      | 91.8 $\pm$ 2.2  | 92.5 $\pm$ 3.2 | 95.1 $\pm$ 1.8  | 81 [10–20]     | 10 [12]     | 653 [10–144] |
| TS                 | 93.9 $\pm$ 1.9      | 93.8 $\pm$ 1.0  | 92.4 $\pm$ 4.1 | 92.9 $\pm$ 3.4  | 80 [10–20]     | 10 [12]     | 270 [10–144] |

Table 21: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for  $P = 7$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. Number of instances are shown in green, with the range of graph sizes shown in brackets.

## 8 Tables for depth $P = 8$

|                    | ER         | HH          | L2F          | RR          |
|--------------------|------------|-------------|--------------|-------------|
| MPS                |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 120 [40–100] | 21 [40–100] |
| Fixed Angle (Aer)* | -          | -           | -            | 10 [40]     |
| Fixed Angle        | -          | -           | 10 [100]     | 10 [40]     |
| Fixed Angle*       | -          | -           | 10 [100]     | 10 [40]     |
| Fourier (Aer)      | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144] | 120 [40–100] | 21 [40–100] |
| Interp (Aer)       | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| Linear Ramp (Aer)* | -          | -           | -            | -           |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA (Aer)*         | 10 [40]    | 10 [39]     | 10 [40]      | 10 [40]     |
| TQA                | 20 [40–50] | 10 [39]     | 120 [40–100] | 20 [40]     |
| TQA*               | 20 [40–50] | 10 [39]     | 120 [40–100] | 20 [40]     |
| PP                 |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 120 [40–100] | 21 [40–100] |
| Fixed Angle        | -          | -           | 10 [100]     | 10 [40]     |
| Fixed Angle*       | -          | -           | 10 [100]     | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144] | 120 [40–100] | 21 [40–100] |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA                | 20 [40–50] | 10 [39]     | 120 [40–100] | 20 [40]     |
| TQA*               | 20 [40–50] | 10 [39]     | 120 [40–100] | 20 [40]     |
| SV                 |            |             |              |             |
| Fourier            | 53 [10–18] | 10 [12]     | 681 [10–20]  | 260 [10–20] |
| Fixed Angle        | 16 [11–17] | -           | 18 [10–18]   | 48 [10–20]  |
| Fixed Angle*       | 16 [11–17] | -           | 18 [10–18]   | 48 [10–20]  |
| Interp             | 82 [10–20] | 10 [12]     | 676 [10–20]  | 370 [10–20] |
| Linear Ramp*       | -          | -           | -            | -           |
| TQA                | 83 [10–20] | 10 [12]     | 659 [10–20]  | 353 [10–20] |
| TQA*               | 81 [10–20] | 10 [12]     | 653 [10–20]  | 132 [10–20] |
| TS                 | 80 [10–20] | 10 [12]     | 265 [10–20]  | 348 [10–20] |

Table 22: **Number of Instances (num. nodes in brackets) for  $P = 8$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                    | ER              | HH              | L2F             | RR              |
|--------------------|-----------------|-----------------|-----------------|-----------------|
| MPS                |                 |                 |                 |                 |
| Fourier            | 81.3 $\pm$ 6.6  | 76.8 $\pm$ 10.2 | 88.2 $\pm$ 5.2  | 93.8 $\pm$ 6.3  |
| Fixed Angle (Aer)* | -               | -               | -               | 98.4 $\pm$ 0.5  |
| Fixed Angle        | -               | -               | 96.6 $\pm$ 0.8  | 97.9 $\pm$ 0.4  |
| Fixed Angle*       | -               | -               | 96.7 $\pm$ 0.8  | 98.3 $\pm$ 0.3  |
| Fourier (Aer)      | 79.5 $\pm$ 7.1  | 87.3 $\pm$ 3.2  | 90.4 $\pm$ 4.4  | 86.5 $\pm$ 8.7  |
| Interp             | 96.9 $\pm$ 1.3  | 84.3 $\pm$ 10.1 | 95.6 $\pm$ 1.0  | 96.1 $\pm$ 3.2  |
| Interp (Aer)       | 94.3 $\pm$ 1.5  | 94.4 $\pm$ 1.1  | 97.2 $\pm$ 0.6  | 90.8 $\pm$ 2.7  |
| Linear Ramp (Aer)* | -               | -               | -               | -               |
| Linear Ramp*       | -               | -               | -               | -               |
| Recursive TS       | -               | -               | -               | -               |
| TQA (Aer)*         | 86.3 $\pm$ 1.7  | 89.3 $\pm$ 1.0  | 93.3 $\pm$ 1.5  | 96.0 $\pm$ 0.5  |
| TQA                | 82.2 $\pm$ 6.9  | 88.3 $\pm$ 1.2  | 89.3 $\pm$ 2.0  | 93.5 $\pm$ 2.0  |
| TQA*               | 91.1 $\pm$ 3.7  | 90.3 $\pm$ 1.0  | 92.2 $\pm$ 1.2  | 96.4 $\pm$ 1.0  |
| PP                 |                 |                 |                 |                 |
| Fourier            | 91.0 $\pm$ 1.9  | 90.1 $\pm$ 3.0  | 85.7 $\pm$ 2.0  | 90.4 $\pm$ 2.2  |
| Fixed Angle        | -               | -               | 86.2 $\pm$ 1.7  | 92.8 $\pm$ 1.6  |
| Fixed Angle*       | -               | -               | 90.3 $\pm$ 1.9  | 93.4 $\pm$ 1.6  |
| Interp             | 93.7 $\pm$ 2.0  | 93.9 $\pm$ 0.9  | 88.8 $\pm$ 1.8  | 91.9 $\pm$ 2.1  |
| Linear Ramp*       | -               | -               | -               | -               |
| Recursive TS       | -               | -               | -               | -               |
| TQA                | 84.6 $\pm$ 2.2  | 88.2 $\pm$ 1.2  | 77.4 $\pm$ 6.7  | 89.9 $\pm$ 1.6  |
| TQA*               | 91.7 $\pm$ 2.3  | 91.1 $\pm$ 0.7  | 85.6 $\pm$ 2.5  | 92.0 $\pm$ 1.5  |
| SV                 |                 |                 |                 |                 |
| Fourier            | 79.0 $\pm$ 12.3 | 79.4 $\pm$ 11.1 | 79.7 $\pm$ 11.2 | 84.4 $\pm$ 12.7 |
| Fixed Angle        | 91.8 $\pm$ 3.2  | -               | 96.8 $\pm$ 1.1  | 97.3 $\pm$ 1.1  |
| Fixed Angle*       | 96.4 $\pm$ 1.4  | -               | 98.1 $\pm$ 0.9  | 97.8 $\pm$ 1.2  |
| Interp             | 95.8 $\pm$ 1.2  | 94.7 $\pm$ 1.1  | 95.3 $\pm$ 4.0  | 93.7 $\pm$ 5.4  |
| Linear Ramp*       | -               | -               | -               | -               |
| TQA                | 90.2 $\pm$ 3.5  | 88.1 $\pm$ 4.0  | 83.5 $\pm$ 8.5  | 90.0 $\pm$ 4.9  |
| TQA*               | 94.4 $\pm$ 1.3  | 92.3 $\pm$ 1.8  | 93.2 $\pm$ 3.3  | 95.7 $\pm$ 1.7  |
| TS                 | 94.6 $\pm$ 1.8  | 94.3 $\pm$ 1.0  | 93.3 $\pm$ 4.6  | 93.4 $\pm$ 3.3  |

Table 23: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for  $P = 8$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|                    | Approximation Ratio |                 |                 |                 | Num. Instances |             |              |
|--------------------|---------------------|-----------------|-----------------|-----------------|----------------|-------------|--------------|
|                    | ER                  | HH              | L2F             | RR              | ER             | HH          | L2F          |
| MPS                |                     |                 |                 |                 |                |             |              |
| Fourier            | 81.3 $\pm$ 6.6      | 76.8 $\pm$ 10.2 | 88.2 $\pm$ 5.2  | 93.8 $\pm$ 6.3  | 20 [40–50]     | 31 [39–144] | 120 [40–144] |
| Fixed Angle (Aer)* | -                   | -               | -               | 98.4 $\pm$ 0.5  | -              | -           | -            |
| Fixed Angle        | -                   | -               | 96.6 $\pm$ 0.8  | 97.9 $\pm$ 0.4  | -              | -           | 10 [10–144]  |
| Fixed Angle*       | -                   | -               | 96.7 $\pm$ 0.8  | 98.3 $\pm$ 0.3  | -              | -           | 10 [10–144]  |
| Fourier (Aer)      | 79.5 $\pm$ 7.1      | 87.3 $\pm$ 3.2  | 90.4 $\pm$ 4.4  | 86.5 $\pm$ 8.7  | 10 [40]        | 10 [39]     | 10 [40–144]  |
| Interp             | 96.9 $\pm$ 1.3      | 84.3 $\pm$ 10.1 | 95.6 $\pm$ 1.0  | 96.1 $\pm$ 3.2  | 20 [40–50]     | 31 [39–144] | 120 [40–144] |
| Interp (Aer)       | 94.3 $\pm$ 1.5      | 94.4 $\pm$ 1.1  | 97.2 $\pm$ 0.6  | 90.8 $\pm$ 2.7  | 10 [40]        | 10 [39]     | 10 [40–144]  |
| Linear Ramp (Aer)* | -                   | -               | -               | -               | -              | -           | -            |
| Linear Ramp*       | -                   | -               | -               | -               | -              | -           | -            |
| Recursive TS       | -                   | -               | -               | -               | -              | -           | -            |
| TQA (Aer)*         | 86.3 $\pm$ 1.7      | 89.3 $\pm$ 1.0  | 93.3 $\pm$ 1.5  | 96.0 $\pm$ 0.5  | 10 [40]        | 10 [39]     | 10 [40–144]  |
| TQA                | 82.2 $\pm$ 6.9      | 88.3 $\pm$ 1.2  | 89.3 $\pm$ 2.0  | 93.5 $\pm$ 2.0  | 20 [40–50]     | 10 [39]     | 120 [40–144] |
| TQA*               | 91.1 $\pm$ 3.7      | 90.3 $\pm$ 1.0  | 92.2 $\pm$ 1.2  | 96.4 $\pm$ 1.0  | 20 [40–50]     | 10 [39]     | 120 [40–144] |
| PP                 |                     |                 |                 |                 |                |             |              |
| Fourier            | 91.0 $\pm$ 1.9      | 90.1 $\pm$ 3.0  | 85.7 $\pm$ 2.0  | 90.4 $\pm$ 2.2  | 20 [40–50]     | 31 [39–144] | 120 [40–144] |
| Fixed Angle        | -                   | -               | 86.2 $\pm$ 1.7  | 92.8 $\pm$ 1.6  | -              | -           | 10 [10–144]  |
| Fixed Angle*       | -                   | -               | 90.3 $\pm$ 1.9  | 93.4 $\pm$ 1.6  | -              | -           | 10 [10–144]  |
| Interp             | 93.7 $\pm$ 2.0      | 93.9 $\pm$ 0.9  | 88.8 $\pm$ 1.8  | 91.9 $\pm$ 2.1  | 20 [40–50]     | 31 [39–144] | 120 [40–144] |
| Linear Ramp*       | -                   | -               | -               | -               | -              | -           | -            |
| Recursive TS       | -                   | -               | -               | -               | -              | -           | -            |
| TQA                | 84.6 $\pm$ 2.2      | 88.2 $\pm$ 1.2  | 77.4 $\pm$ 6.7  | 89.9 $\pm$ 1.6  | 20 [40–50]     | 10 [39]     | 120 [40–144] |
| TQA*               | 91.7 $\pm$ 2.3      | 91.1 $\pm$ 0.7  | 85.6 $\pm$ 2.5  | 92.0 $\pm$ 1.5  | 20 [40–50]     | 10 [39]     | 120 [40–144] |
| SV                 |                     |                 |                 |                 |                |             |              |
| Fourier            | 79.0 $\pm$ 12.3     | 79.4 $\pm$ 11.1 | 79.7 $\pm$ 11.2 | 84.4 $\pm$ 12.7 | 53 [10–18]     | 10 [12]     | 681 [10–144] |
| Fixed Angle        | 91.8 $\pm$ 3.2      | -               | 96.8 $\pm$ 1.1  | 97.3 $\pm$ 1.1  | 16 [11–17]     | -           | 18 [10–144]  |
| Fixed Angle*       | 96.4 $\pm$ 1.4      | -               | 98.1 $\pm$ 0.9  | 97.8 $\pm$ 1.2  | 16 [11–17]     | -           | 18 [10–144]  |
| Interp             | 95.8 $\pm$ 1.2      | 94.7 $\pm$ 1.1  | 95.3 $\pm$ 4.0  | 93.7 $\pm$ 5.4  | 82 [10–20]     | 10 [12]     | 676 [10–144] |
| Linear Ramp*       | -                   | -               | -               | -               | -              | -           | -            |
| TQA                | 90.2 $\pm$ 3.5      | 88.1 $\pm$ 4.0  | 83.5 $\pm$ 8.5  | 90.0 $\pm$ 4.9  | 83 [10–20]     | 10 [12]     | 659 [10–144] |
| TQA*               | 94.4 $\pm$ 1.3      | 92.3 $\pm$ 1.8  | 93.2 $\pm$ 3.3  | 95.7 $\pm$ 1.7  | 81 [10–20]     | 10 [12]     | 653 [10–144] |
| TS                 | 94.6 $\pm$ 1.8      | 94.3 $\pm$ 1.0  | 93.3 $\pm$ 4.6  | 93.4 $\pm$ 3.3  | 80 [10–20]     | 10 [12]     | 265 [10–144] |

Table 24: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for  $P = 8$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. Number of instances are shown in green, with the range of graph sizes shown in brackets.



## 9 Tables for depth $P = 9$

|                    | ER         | HH          | L2F          | RR          |
|--------------------|------------|-------------|--------------|-------------|
| MPS                |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 111 [40–100] | 11 [40–100] |
| Fixed Angle (Aer)* | -          | -           | -            | -           |
| Fixed Angle        | -          | -           | -            | -           |
| Fixed Angle*       | -          | -           | -            | -           |
| Fourier (Aer)      | 10 [40]    | 10 [39]     | -            | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144] | 111 [40–100] | 11 [40–100] |
| Interp (Aer)       | 10 [40]    | 10 [39]     | -            | 10 [40]     |
| Linear Ramp (Aer)* | -          | -           | -            | -           |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA (Aer)*         | -          | -           | -            | -           |
| TQA                | -          | -           | -            | -           |
| TQA*               | -          | -           | -            | -           |
| PP                 |            |             |              |             |
| Fourier            | 20 [40–50] | 31 [39–144] | 111 [40–100] | 11 [40–100] |
| Fixed Angle        | -          | -           | -            | -           |
| Fixed Angle*       | -          | -           | -            | -           |
| Interp             | 20 [40–50] | 31 [39–144] | 111 [40–100] | 11 [40–100] |
| Linear Ramp*       | -          | -           | -            | -           |
| Recursive TS       | -          | -           | -            | -           |
| TQA                | -          | -           | -            | -           |
| TQA*               | -          | -           | -            | -           |
| SV                 |            |             |              |             |
| Fourier            | 52 [10–18] | 10 [12]     | 672 [10–20]  | 258 [10–20] |
| Fixed Angle        | 16 [11–17] | -           | 18 [10–18]   | 48 [10–20]  |
| Fixed Angle*       | 16 [11–17] | -           | 18 [10–18]   | 48 [10–20]  |
| Interp             | 82 [10–20] | 10 [12]     | 673 [10–20]  | 369 [10–20] |
| Linear Ramp*       | -          | -           | -            | -           |
| TQA                | 83 [10–20] | 10 [12]     | 658 [10–20]  | 354 [10–20] |
| TQA*               | 81 [10–20] | 10 [12]     | 652 [10–20]  | 131 [10–20] |
| TS                 | 80 [10–20] | 10 [12]     | 264 [10–20]  | 352 [10–20] |

Table 25: **Number of Instances (num. nodes in brackets) for  $P = 9$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                    | ER              | HH              | L2F             | RR              |
|--------------------|-----------------|-----------------|-----------------|-----------------|
| MPS                |                 |                 |                 |                 |
| Fourier            | 82.0 $\pm$ 7.8  | 78.3 $\pm$ 9.7  | 88.6 $\pm$ 4.3  | 92.7 $\pm$ 7.4  |
| Fixed Angle (Aer)* | -               | -               | -               | -               |
| Fixed Angle        | -               | -               | -               | -               |
| Fixed Angle*       | -               | -               | -               | -               |
| Fourier (Aer)      | 79.0 $\pm$ 6.7  | 88.7 $\pm$ 3.6  | -               | 89.7 $\pm$ 3.2  |
| Interp             | 97.2 $\pm$ 1.4  | 86.2 $\pm$ 9.3  | 96.2 $\pm$ 0.9  | 93.1 $\pm$ 4.5  |
| Interp (Aer)       | 94.1 $\pm$ 4.2  | 95.0 $\pm$ 1.0  | -               | 93.5 $\pm$ 1.9  |
| Linear Ramp (Aer)* | -               | -               | -               | -               |
| Linear Ramp*       | -               | -               | -               | -               |
| Recursive TS       | -               | -               | -               | -               |
| TQA (Aer)*         | -               | -               | -               | -               |
| TQA                | -               | -               | -               | -               |
| TQA*               | -               | -               | -               | -               |
| PP                 |                 |                 |                 |                 |
| Fourier            | 91.7 $\pm$ 1.6  | 90.6 $\pm$ 3.2  | 86.2 $\pm$ 1.9  | 91.0 $\pm$ 2.6  |
| Fixed Angle        | -               | -               | -               | -               |
| Fixed Angle*       | -               | -               | -               | -               |
| Interp             | 94.1 $\pm$ 2.0  | 94.4 $\pm$ 0.8  | 89.2 $\pm$ 1.7  | 90.8 $\pm$ 2.3  |
| Linear Ramp*       | -               | -               | -               | -               |
| Recursive TS       | -               | -               | -               | -               |
| TQA                | -               | -               | -               | -               |
| TQA*               | -               | -               | -               | -               |
| SV                 |                 |                 |                 |                 |
| Fourier            | 78.2 $\pm$ 12.7 | 78.0 $\pm$ 10.8 | 79.1 $\pm$ 11.6 | 83.9 $\pm$ 13.5 |
| Fixed Angle        | 92.1 $\pm$ 3.4  | -               | 97.0 $\pm$ 1.3  | 97.7 $\pm$ 1.1  |
| Fixed Angle*       | 97.0 $\pm$ 1.4  | -               | 98.5 $\pm$ 0.8  | 98.3 $\pm$ 1.1  |
| Interp             | 96.3 $\pm$ 1.1  | 94.8 $\pm$ 1.2  | 95.7 $\pm$ 4.3  | 93.6 $\pm$ 6.2  |
| Linear Ramp*       | -               | -               | -               | -               |
| TQA                | 90.9 $\pm$ 3.7  | 89.1 $\pm$ 3.8  | 83.7 $\pm$ 9.2  | 90.8 $\pm$ 4.9  |
| TQA*               | 95.0 $\pm$ 1.3  | 93.0 $\pm$ 1.6  | 93.6 $\pm$ 3.3  | 96.2 $\pm$ 1.7  |
| TS                 | 95.0 $\pm$ 1.9  | 94.7 $\pm$ 0.9  | 93.6 $\pm$ 5.2  | 93.6 $\pm$ 3.2  |

Table 26: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for  $P = 9$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|                    | Approximation Ratio |                 |                 |                 | Num. Instances |             |          |
|--------------------|---------------------|-----------------|-----------------|-----------------|----------------|-------------|----------|
|                    | ER                  | HH              | L2F             | RR              | ER             | HH          | L2F      |
| MPS                |                     |                 |                 |                 |                |             |          |
| Fourier            | 82.0 $\pm$ 7.8      | 78.3 $\pm$ 9.7  | 88.6 $\pm$ 4.3  | 92.7 $\pm$ 7.4  | 20 [40–50]     | 31 [39–144] | 111 [40– |
| Fixed Angle (Aer)* | -                   | -               | -               | -               | -              | -           | -        |
| Fixed Angle        | -                   | -               | -               | -               | -              | -           | -        |
| Fixed Angle*       | -                   | -               | -               | -               | -              | -           | -        |
| Fourier (Aer)      | 79.0 $\pm$ 6.7      | 88.7 $\pm$ 3.6  | -               | 89.7 $\pm$ 3.2  | 10 [40]        | 10 [39]     | -        |
| Interp             | 97.2 $\pm$ 1.4      | 86.2 $\pm$ 9.3  | 96.2 $\pm$ 0.9  | 93.1 $\pm$ 4.5  | 20 [40–50]     | 31 [39–144] | 111 [40– |
| Interp (Aer)       | 94.1 $\pm$ 4.2      | 95.0 $\pm$ 1.0  | -               | 93.5 $\pm$ 1.9  | 10 [40]        | 10 [39]     | -        |
| Linear Ramp (Aer)* | -                   | -               | -               | -               | -              | -           | -        |
| Linear Ramp*       | -                   | -               | -               | -               | -              | -           | -        |
| Recursive TS       | -                   | -               | -               | -               | -              | -           | -        |
| TQA (Aer)*         | -                   | -               | -               | -               | -              | -           | -        |
| TQA                | -                   | -               | -               | -               | -              | -           | -        |
| TQA*               | -                   | -               | -               | -               | -              | -           | -        |
| PP                 |                     |                 |                 |                 |                |             |          |
| Fourier            | 91.7 $\pm$ 1.6      | 90.6 $\pm$ 3.2  | 86.2 $\pm$ 1.9  | 91.0 $\pm$ 2.6  | 20 [40–50]     | 31 [39–144] | 111 [40– |
| Fixed Angle        | -                   | -               | -               | -               | -              | -           | -        |
| Fixed Angle*       | -                   | -               | -               | -               | -              | -           | -        |
| Interp             | 94.1 $\pm$ 2.0      | 94.4 $\pm$ 0.8  | 89.2 $\pm$ 1.7  | 90.8 $\pm$ 2.3  | 20 [40–50]     | 31 [39–144] | 111 [40– |
| Linear Ramp*       | -                   | -               | -               | -               | -              | -           | -        |
| Recursive TS       | -                   | -               | -               | -               | -              | -           | -        |
| TQA                | -                   | -               | -               | -               | -              | -           | -        |
| TQA*               | -                   | -               | -               | -               | -              | -           | -        |
| SV                 |                     |                 |                 |                 |                |             |          |
| Fourier            | 78.2 $\pm$ 12.7     | 78.0 $\pm$ 10.8 | 79.1 $\pm$ 11.6 | 83.9 $\pm$ 13.5 | 52 [10–18]     | 10 [12]     | 672 [10– |
| Fixed Angle        | 92.1 $\pm$ 3.4      | -               | 97.0 $\pm$ 1.3  | 97.7 $\pm$ 1.1  | 16 [11–17]     | -           | 18 [10–  |
| Fixed Angle*       | 97.0 $\pm$ 1.4      | -               | 98.5 $\pm$ 0.8  | 98.3 $\pm$ 1.1  | 16 [11–17]     | -           | 18 [10–  |
| Interp             | 96.3 $\pm$ 1.1      | 94.8 $\pm$ 1.2  | 95.7 $\pm$ 4.3  | 93.6 $\pm$ 6.2  | 82 [10–20]     | 10 [12]     | 673 [10– |
| Linear Ramp*       | -                   | -               | -               | -               | -              | -           | -        |
| TQA                | 90.9 $\pm$ 3.7      | 89.1 $\pm$ 3.8  | 83.7 $\pm$ 9.2  | 90.8 $\pm$ 4.9  | 83 [10–20]     | 10 [12]     | 658 [10– |
| TQA*               | 95.0 $\pm$ 1.3      | 93.0 $\pm$ 1.6  | 93.6 $\pm$ 3.3  | 96.2 $\pm$ 1.7  | 81 [10–20]     | 10 [12]     | 652 [10– |
| TS                 | 95.0 $\pm$ 1.9      | 94.7 $\pm$ 0.9  | 93.6 $\pm$ 5.2  | 93.6 $\pm$ 3.2  | 80 [10–20]     | 10 [12]     | 264 [10– |

Table 27: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for  $P = 9$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. Number of instances are shown in green, with the range of graph sizes shown in brackets.

## 10 Tables for depth $P = 10$

|                    | ER         | HH           | L2F          | RR          |
|--------------------|------------|--------------|--------------|-------------|
| MPS                |            |              |              |             |
| Fourier            | 20 [40–50] | 31 [39–144]  | 111 [40–100] | 11 [40–100] |
| Fixed Angle (Aer)* | -          | -            | -            | 10 [40]     |
| Fixed Angle        | -          | -            | 10 [100]     | 11 [40–100] |
| Fixed Angle*       | -          | -            | 10 [100]     | 11 [40–100] |
| Fourier (Aer)      | 10 [40]    | 10 [39]      | -            | 10 [40]     |
| Interp             | 20 [40–50] | 31 [39–144]  | 111 [40–100] | 11 [40–100] |
| Interp (Aer)       | 10 [40]    | 10 [39]      | -            | 10 [40]     |
| Linear Ramp (Aer)* | -          | 20 [105–144] | 37 [40–100]  | -           |
| Linear Ramp*       | 10 [40]    | 10 [39]      | -            | -           |
| Recursive TS       | -          | -            | -            | -           |
| TQA (Aer)*         | 10 [40]    | 10 [39]      | -            | 10 [40]     |
| TQA                | 20 [40–50] | 31 [39–144]  | 111 [40–100] | 11 [40–100] |
| TQA*               | 20 [40–50] | 31 [39–144]  | 111 [40–100] | 11 [40–100] |
| PP                 |            |              |              |             |
| Fourier            | 20 [40–50] | 31 [39–144]  | 111 [40–100] | 11 [40–100] |
| Fixed Angle        | -          | -            | 10 [100]     | 11 [40–100] |
| Fixed Angle*       | -          | -            | 10 [100]     | 11 [40–100] |
| Interp             | 20 [40–50] | 31 [39–144]  | 111 [40–100] | 11 [40–100] |
| Linear Ramp*       | 10 [40]    | 10 [39]      | 118 [40–100] | -           |
| Recursive TS       | -          | -            | -            | -           |
| TQA                | 20 [40–50] | 30 [39–144]  | 111 [40–100] | 11 [40–100] |
| TQA*               | 20 [40–50] | 30 [39–144]  | 111 [40–100] | 11 [40–100] |
| SV                 |            |              |              |             |
| Fourier            | 50 [10–17] | 10 [12]      | 665 [10–20]  | 245 [10–20] |
| Fixed Angle        | 16 [11–17] | -            | 18 [10–18]   | 42 [10–24]  |
| Fixed Angle*       | 16 [11–17] | -            | 18 [10–18]   | 39 [10–22]  |
| Interp             | 82 [10–20] | 10 [12]      | 664 [10–20]  | 348 [10–20] |
| Linear Ramp*       | 88 [10–20] | 10 [12]      | 175 [10–20]  | 568 [10–20] |
| TQA                | 83 [10–20] | 10 [12]      | 657 [10–20]  | 345 [10–22] |
| TQA*               | 81 [10–20] | 9 [12]       | 651 [10–20]  | 139 [10–20] |
| TS                 | 79 [10–20] | 9 [12]       | 264 [10–20]  | 333 [10–20] |

Table 28: **Number of Instances (num. nodes in brackets) for  $P = 10$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                    | ER              | HH              | L2F             | RR              |
|--------------------|-----------------|-----------------|-----------------|-----------------|
| MPS                |                 |                 |                 |                 |
| Fourier            | 79.6 $\pm$ 6.1  | 77.4 $\pm$ 10.9 | 88.7 $\pm$ 5.6  | 92.8 $\pm$ 9.0  |
| Fixed Angle (Aer)* | -               | -               | -               | 99.0 $\pm$ 0.4  |
| Fixed Angle        | -               | -               | 97.4 $\pm$ 0.7  | 98.2 $\pm$ 1.4  |
| Fixed Angle*       | -               | -               | 97.6 $\pm$ 0.7  | 98.6 $\pm$ 0.9  |
| Fourier (Aer)      | 82.5 $\pm$ 7.8  | 87.7 $\pm$ 5.2  | -               | 91.6 $\pm$ 3.7  |
| Interp             | 97.4 $\pm$ 1.5  | 86.6 $\pm$ 10.1 | 96.6 $\pm$ 0.9  | 93.8 $\pm$ 4.6  |
| Interp (Aer)       | 94.3 $\pm$ 4.5  | 95.6 $\pm$ 0.8  | -               | 92.0 $\pm$ 1.6  |
| Linear Ramp (Aer)* | -               | 87.8 $\pm$ 4.5  | 91.7 $\pm$ 3.8  | -               |
| Linear Ramp*       | 96.8 $\pm$ 0.5  | 93.6 $\pm$ 0.7  | -               | -               |
| Recursive TS       | -               | -               | -               | -               |
| TQA (Aer)*         | 90.1 $\pm$ 1.9  | 90.6 $\pm$ 0.9  | -               | 96.5 $\pm$ 0.6  |
| TQA                | 87.7 $\pm$ 1.8  | 79.6 $\pm$ 9.2  | 90.0 $\pm$ 2.5  | 94.0 $\pm$ 3.3  |
| TQA*               | 91.8 $\pm$ 2.9  | 80.8 $\pm$ 9.3  | 92.6 $\pm$ 1.7  | 96.5 $\pm$ 1.7  |
| PP                 |                 |                 |                 |                 |
| Fourier            | 91.5 $\pm$ 1.9  | 91.1 $\pm$ 3.2  | 86.0 $\pm$ 2.2  | 91.3 $\pm$ 2.6  |
| Fixed Angle        | -               | -               | 86.7 $\pm$ 1.7  | 93.0 $\pm$ 1.6  |
| Fixed Angle*       | -               | -               | 90.7 $\pm$ 1.9  | 93.6 $\pm$ 1.6  |
| Interp             | 94.4 $\pm$ 1.9  | 94.9 $\pm$ 0.8  | 89.6 $\pm$ 1.7  | 90.1 $\pm$ 2.4  |
| Linear Ramp*       | 94.8 $\pm$ 1.4  | 93.4 $\pm$ 0.7  | 87.6 $\pm$ 1.7  | -               |
| Recursive TS       | -               | -               | -               | -               |
| TQA                | 84.4 $\pm$ 2.4  | 90.0 $\pm$ 0.9  | 77.5 $\pm$ 7.8  | 90.8 $\pm$ 1.6  |
| TQA*               | 91.3 $\pm$ 2.5  | 91.8 $\pm$ 0.7  | 85.6 $\pm$ 3.0  | 92.3 $\pm$ 1.6  |
| SV                 |                 |                 |                 |                 |
| Fourier            | 77.5 $\pm$ 13.2 | 78.5 $\pm$ 11.7 | 77.9 $\pm$ 12.3 | 83.7 $\pm$ 13.8 |
| Fixed Angle        | 92.4 $\pm$ 3.5  | -               | 97.2 $\pm$ 1.3  | 98.1 $\pm$ 0.9  |
| Fixed Angle*       | 97.4 $\pm$ 1.4  | -               | 98.8 $\pm$ 0.6  | 98.7 $\pm$ 0.9  |
| Interp             | 96.6 $\pm$ 1.1  | 94.8 $\pm$ 1.1  | 96.0 $\pm$ 3.9  | 94.1 $\pm$ 5.8  |
| Linear Ramp*       | 95.0 $\pm$ 1.5  | 94.5 $\pm$ 1.4  | 93.5 $\pm$ 2.0  | 96.2 $\pm$ 1.8  |
| TQA                | 91.3 $\pm$ 3.9  | 89.9 $\pm$ 3.6  | 83.8 $\pm$ 9.5  | 91.0 $\pm$ 5.1  |
| TQA*               | 95.4 $\pm$ 1.4  | 93.2 $\pm$ 2.1  | 93.9 $\pm$ 3.2  | 96.8 $\pm$ 1.5  |
| TS                 | 95.3 $\pm$ 2.0  | 95.2 $\pm$ 0.8  | 93.4 $\pm$ 4.4  | 94.0 $\pm$ 3.2  |

Table 29: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for  $P = 10$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|                    | Approximation Ratio |                 |                 |                 | Num. Instances |              |              |
|--------------------|---------------------|-----------------|-----------------|-----------------|----------------|--------------|--------------|
|                    | ER                  | HH              | L2F             | RR              | ER             | HH           | L2F          |
| MPS                |                     |                 |                 |                 |                |              |              |
| Fourier            | 79.6 $\pm$ 6.1      | 77.4 $\pm$ 10.9 | 88.7 $\pm$ 5.6  | 92.8 $\pm$ 9.0  | 20 [40–50]     | 31 [39–144]  | 111 [40–144] |
| Fixed Angle (Aer)* | -                   | -               | -               | 99.0 $\pm$ 0.4  | -              | -            | -            |
| Fixed Angle        | -                   | -               | 97.4 $\pm$ 0.7  | 98.2 $\pm$ 1.4  | -              | -            | 10 [10–144]  |
| Fixed Angle*       | -                   | -               | 97.6 $\pm$ 0.7  | 98.6 $\pm$ 0.9  | -              | -            | 10 [10–144]  |
| Fourier (Aer)      | 82.5 $\pm$ 7.8      | 87.7 $\pm$ 5.2  | -               | 91.6 $\pm$ 3.7  | 10 [40]        | 10 [39]      | -            |
| Interp             | 97.4 $\pm$ 1.5      | 86.6 $\pm$ 10.1 | 96.6 $\pm$ 0.9  | 93.8 $\pm$ 4.6  | 20 [40–50]     | 31 [39–144]  | 111 [40–144] |
| Interp (Aer)       | 94.3 $\pm$ 4.5      | 95.6 $\pm$ 0.8  | -               | 92.0 $\pm$ 1.6  | 10 [40]        | 10 [39]      | -            |
| Linear Ramp (Aer)* | -                   | 87.8 $\pm$ 4.5  | 91.7 $\pm$ 3.8  | -               | -              | 20 [105–144] | 37 [40–144]  |
| Linear Ramp*       | 96.8 $\pm$ 0.5      | 93.6 $\pm$ 0.7  | -               | -               | 10 [40]        | 10 [39]      | -            |
| Recursive TS       | -                   | -               | -               | -               | -              | -            | -            |
| TQA (Aer)*         | 90.1 $\pm$ 1.9      | 90.6 $\pm$ 0.9  | -               | 96.5 $\pm$ 0.6  | 10 [40]        | 10 [39]      | -            |
| TQA                | 87.7 $\pm$ 1.8      | 79.6 $\pm$ 9.2  | 90.0 $\pm$ 2.5  | 94.0 $\pm$ 3.3  | 20 [40–50]     | 31 [39–144]  | 111 [40–144] |
| TQA*               | 91.8 $\pm$ 2.9      | 80.8 $\pm$ 9.3  | 92.6 $\pm$ 1.7  | 96.5 $\pm$ 1.7  | 20 [40–50]     | 31 [39–144]  | 111 [40–144] |
| PP                 |                     |                 |                 |                 |                |              |              |
| Fourier            | 91.5 $\pm$ 1.9      | 91.1 $\pm$ 3.2  | 86.0 $\pm$ 2.2  | 91.3 $\pm$ 2.6  | 20 [40–50]     | 31 [39–144]  | 111 [40–144] |
| Fixed Angle        | -                   | -               | 86.7 $\pm$ 1.7  | 93.0 $\pm$ 1.6  | -              | -            | 10 [10–144]  |
| Fixed Angle*       | -                   | -               | 90.7 $\pm$ 1.9  | 93.6 $\pm$ 1.6  | -              | -            | 10 [10–144]  |
| Interp             | 94.4 $\pm$ 1.9      | 94.9 $\pm$ 0.8  | 89.6 $\pm$ 1.7  | 90.1 $\pm$ 2.4  | 20 [40–50]     | 31 [39–144]  | 111 [40–144] |
| Linear Ramp*       | 94.8 $\pm$ 1.4      | 93.4 $\pm$ 0.7  | 87.6 $\pm$ 1.7  | -               | 10 [40]        | 10 [39]      | 118 [40–144] |
| Recursive TS       | -                   | -               | -               | -               | -              | -            | -            |
| TQA                | 84.4 $\pm$ 2.4      | 90.0 $\pm$ 0.9  | 77.5 $\pm$ 7.8  | 90.8 $\pm$ 1.6  | 20 [40–50]     | 30 [39–144]  | 111 [40–144] |
| TQA*               | 91.3 $\pm$ 2.5      | 91.8 $\pm$ 0.7  | 85.6 $\pm$ 3.0  | 92.3 $\pm$ 1.6  | 20 [40–50]     | 30 [39–144]  | 111 [40–144] |
| SV                 |                     |                 |                 |                 |                |              |              |
| Fourier            | 77.5 $\pm$ 13.2     | 78.5 $\pm$ 11.7 | 77.9 $\pm$ 12.3 | 83.7 $\pm$ 13.8 | 50 [10–17]     | 10 [12]      | 665 [10–144] |
| Fixed Angle        | 92.4 $\pm$ 3.5      | -               | 97.2 $\pm$ 1.3  | 98.1 $\pm$ 0.9  | 16 [11–17]     | -            | 18 [10–144]  |
| Fixed Angle*       | 97.4 $\pm$ 1.4      | -               | 98.8 $\pm$ 0.6  | 98.7 $\pm$ 0.9  | 16 [11–17]     | -            | 18 [10–144]  |
| Interp             | 96.6 $\pm$ 1.1      | 94.8 $\pm$ 1.1  | 96.0 $\pm$ 3.9  | 94.1 $\pm$ 5.8  | 82 [10–20]     | 10 [12]      | 664 [10–144] |
| Linear Ramp*       | 95.0 $\pm$ 1.5      | 94.5 $\pm$ 1.4  | 93.5 $\pm$ 2.0  | 96.2 $\pm$ 1.8  | 88 [10–20]     | 10 [12]      | 175 [10–144] |
| TQA                | 91.3 $\pm$ 3.9      | 89.9 $\pm$ 3.6  | 83.8 $\pm$ 9.5  | 91.0 $\pm$ 5.1  | 83 [10–20]     | 10 [12]      | 657 [10–144] |
| TQA*               | 95.4 $\pm$ 1.4      | 93.2 $\pm$ 2.1  | 93.9 $\pm$ 3.2  | 96.8 $\pm$ 1.5  | 81 [10–20]     | 9 [12]       | 651 [10–144] |
| TS                 | 95.3 $\pm$ 2.0      | 95.2 $\pm$ 0.8  | 93.4 $\pm$ 4.4  | 94.0 $\pm$ 3.2  | 79 [10–20]     | 9 [12]       | 264 [10–144] |

Table 30: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for  $P = 10$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. Number of instances are shown in green, with the range of graph sizes shown in brackets.